

# GRAPHICS SYSTEM

NAME: \_\_\_\_\_  
DATE: / /

Graphics system def<sup>n</sup>: - combination of hardware & software that is used to create and display graphics

Animation: the process of making <sup>moving picture</sup> objects <sup>object picture</sup> movable by showing them quickly one after another.

IMP  
SN

## Application of Computer Graphics

NAME \_\_\_\_\_  
DATE 11

### ① CAA - (Computer Aided Design)

1. **Engineering Design** – Used to design machines, machine parts, and mechanical components.
2. **Architectural Design** – Used to create building plans, floor plans, and 3D building models.
3. **Automobile Design** – Used to design cars, bikes, engines, and other vehicle components.
4. **Aerospace Design** – Used to design aircraft, spacecraft, and their components.
5. **Electrical/Electronic Design** – Used to create circuit diagrams and PCB layouts.
6. **Product Design** – Used to create and visualize new products before manufacturing.
7. **3D Modeling** – Used to create three-dimensional models and view them from different angles.

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## ② Presentation Graphics

1. presentation graphics are used to show information using picture
2. It is also used to summarize financial, mathematical, statistical, scientific data for research & reports
2. Common examples are:
  - Bar charts
  - Line graph
  - Pie chart
  - Pictorial
  - Surface graph.
3. Graphs and charts are displayed / can be shown using projector or computer screen.
4. Graphs and charts are displayed in 2D/3D for better view & additional information
5. Used in meetings, seminar, research & business reports
6. Helps to understand information quickly
7. used to make charts, graphs, pictorial and diagram for presentation

## ③ Computer Art.

1. Computer Art means creating pictures using computer Graphics
2. Computer graphics methods are used in both line art and commercial art
3. A 'paint' brush program helps to draw pictorial on computer screen.
4. Pictures are drawn ~~using~~ graphic tablets (digital) using a stylus to create different strokes and colors.
5. Artists use 3D modeling, texture mapping, drawing programs & CAD software to create image.
6. Used in advertising, movies, TV animation & game.

## ⑥ Image Processing

1. Image processing means editing and improving images using computer

2. eg is used in image processing also to edit & improve image

3. Helps to improve image quality, brightness & color

4. Used in medical field to see X-ray

CT scan

MRI image clearly

5. Used in satellite images & ~~weather~~ forecasting

6. Used in photo & video editing applications

7. Helps to remove noise, blur, unwanted parts from image

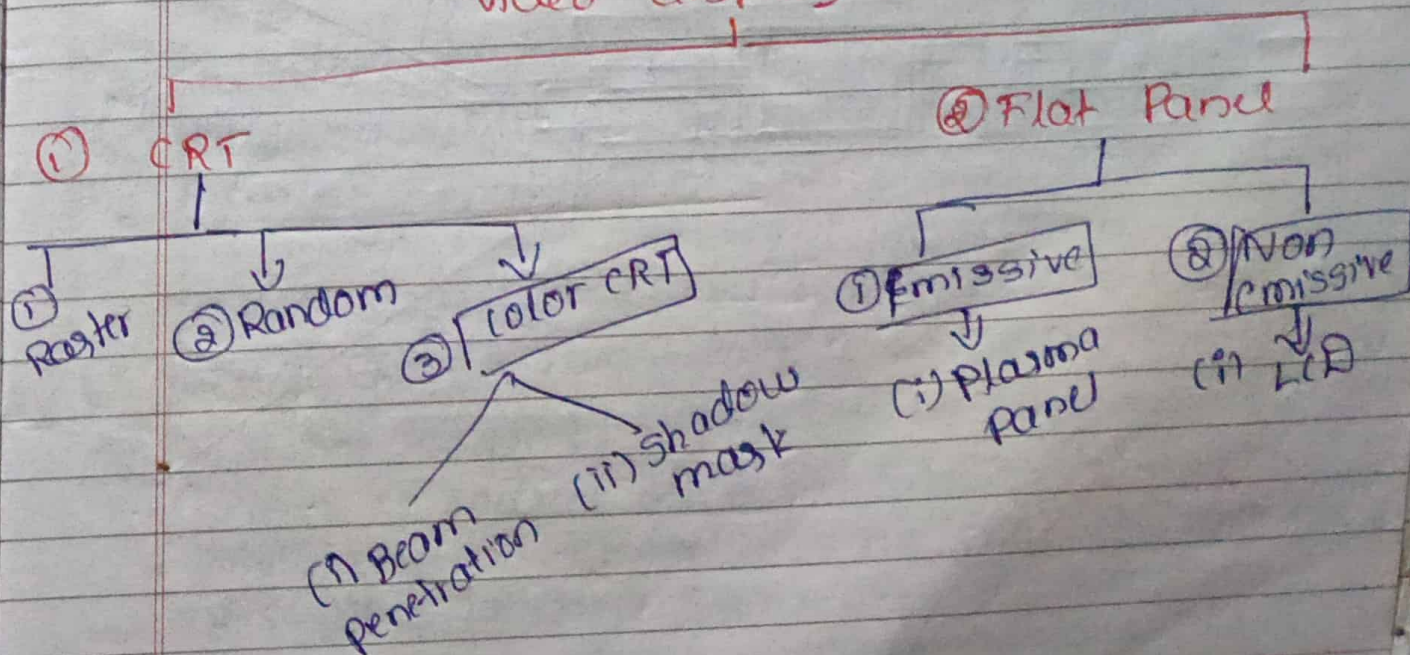
8. Used in science, research & security systems

9. 2 principle of Image Processing

(i) improving picture quality

10. ~~comp~~ (ii) Image processing helps the computer to recognize & understand pictures like in robotics.

## Video display device



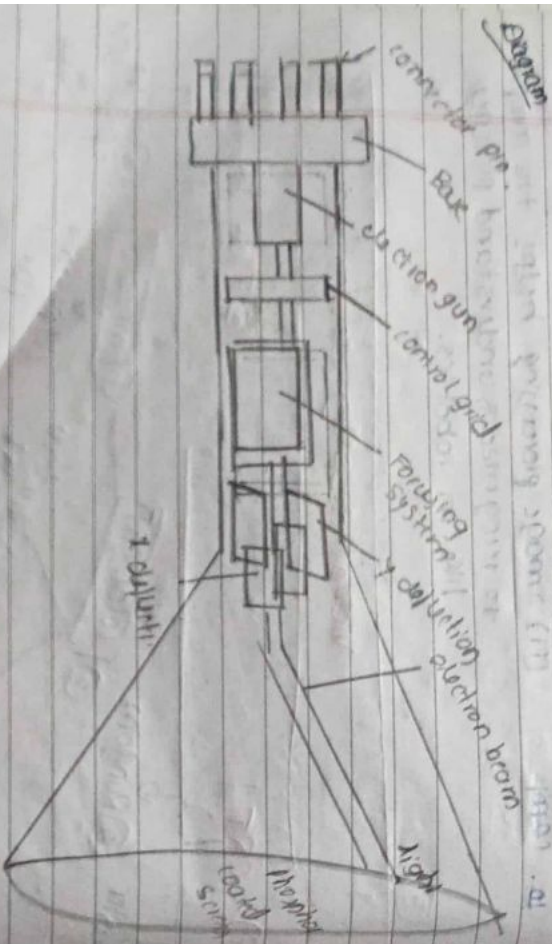
## VIDEO DISPLAY DEVICES

### ① Cathode Ray Tube (CRT)

- ② Cathode Ray Tube is a video display device used in old computers and televisions to show images on screen.
- ③ A CRT works by using an electron beam to draw pictures on screen.

④ CRT is a video display device used to display images on screen by using electron beam & phosphor coated screen.

A CRT is vacuum tube in which images are produced when electron beam strikes a phosphor coated surface.



### Components of CRT

#### ① Vacuum Tube

- The entire assembly (CRT) is enclosed in a vacuum tube.
- The vacuum tube is a glass tube in which the electron beam is produced.
- Air is removed from the tube so that it prevents electrons from colliding with air molecules.
- This helps the electron beam move easily without stopping.

#### ② Electron Gun

- Electron gun is heart of CRT.
- Electron gun produces electrons (beams).
- The beams move towards screen or screen.
- It is used to draw images on screen.
- Electron gun consists of heating filament when filament is heated it produces a very charged electrons.

#### ③ Control Grids

- Control grids are placed near electron gun.
- It controls brightness of the image.
- It controls electrons pass → image is bright.

#### ④ Deflection System

- Deflection system controls the direction of electron beam.
- It moves beam left, right, up, down.
- It uses deflection plates (coils) to move beam.

#### ⑤ Phosphor Coated Screen

- Screen is coated with phosphor.
- When electron beam hits the screen, phosphor glows.
- Because of this light, image appears.
- Different phosphor materials give different colors.

## Operation of CRT / Working Function.

- ① Heating the filament (cathode gun) - Heating filament when the electricity supplied, the filament produces negatively charged electrons. These electrons form electron beam.
- ② Electron gun produces electron beam.
  - Electron gun emits electrons in a narrow beam.
  - Electrons form a thin electron beam.
- ③ Control grid controls brightness/number of electrons.
  - Control grid controls the number of electrons.
  - More electrons → brighter image.
  - Less electrons → dimmer image.
- ④ Focusing system - makes beam clear & sharp.
  - Focusing system makes beam clear & sharp.
  - Beam is focused on correct point on screen.
- ⑤ Deflection system - moves beam up, down, left, right.
  - Deflection system moves beam up, down, left, right.
  - Beam is controlled direction of electron beam.
  - Beam is controlled direction of electron beam.
  - Beam is controlled direction of electron beam.
- ⑥ Phosphor screen - produces light.
  - Screen is coated with phosphor.
  - When electron beam hits phosphor, it glows.
  - This glow produces light.
  - Because of light, image appears on screen.
  - Different phosphor materials give different colors.

- Advantage of CRT
- ① good colour quality
  - ② heavy size
  - ③ high picture consumption
  - ④ high response

Conclusion.  
CRT works using electron beam where electron gun produces electrons, control grid control brightness, deflection system moves beam, phosphor screen glows to produce image. Thus CRT display picture on screen.

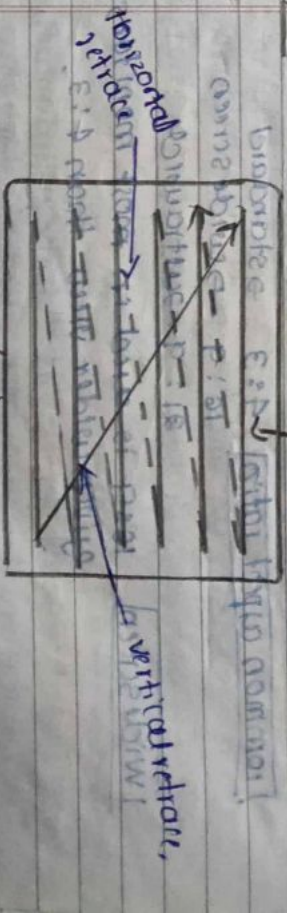
## ⑧ Raster Scan Display

Raster - A rectangular array of points or dots.  
A raster image is a collection of dots called pixels.

### ③ Raster Scan Display

Raster Scan Display is a type of display method system that draws image on screen by line by line (row by row) from the top-left corner to the bottom-right corner using electron beam.  
In raster scan system, refreshing rate is 60-90 frames per second.

refreshing rate described in the unit called Hz (Hertz).  
Diagram



**Raster**  
collection of pixels arranged in rows & columns to form the image on the screen.

# Key concepts / Factors / Parameters of Refresh Scan Display

- ① Resolution
- ② Aspect ratio
- ③ Refresh rate
- ④ Frame

## ① Resolution

Resolution means no. of pixels on screen  
usually represented as width  $\times$  height

eg.  $1920 \times 1080$

more pixels  $\rightarrow$  clearer image.

Pixel density :- means number of pixels in one area of screen

## Formap resolution

Full HD  $1920 \times 1080$   
Quad HD  $2560 \times 1440$   
4K  $3840 \times 2160$

## ② Aspect Ratio

Aspect ratio is the ratio of width to the height of the screen

## Common aspect ratio

4:3  $\rightarrow$  standard  
16:9  $\rightarrow$  wide screen  
21:9  $\rightarrow$  ultra wide

## Widescreen

16:9 is used in most monitors  
give wider view than 4:3.

## 3. Refresh rate

Refresh rate means how many times screen updates in one second

- Measured in Hertz (Hz)
- Higher refresh rate results in smoother picture.
- Low refresh rate results in blur picture.

## Common refresh rate

standard vs. High  
standard refresh rate  $\rightarrow$  60 Hz  
high refresh rate  $\rightarrow$  144 Hz

ports refresh rates which are used in gaming & video application

## ④ Frame

Frame means a complete picture on the screen

Many frames are shown in one second

Frame are used in video & animation.

More frames in a second result in smooth motion

## ⑤ Frame rate

Frame rate is no. of frames displayed per second

## ③ Refresh rate

84 fps  $\rightarrow$  cinema  
90 fps  $\rightarrow$  video  
60 fps  $\rightarrow$  smooth display

## ④ Frame buffer

Frame buffer refresh buffer.  $\rightarrow$  Picture definition is stored in memory area called frame buffer

## ⑤ Horizontal trace

Horizontal trace is the end of each scan line, electron beam again returns to the left side of the screen, begin displaying next scan line.

Horizontal trace is the movement of electron beam from left side to left side after completing one line

vertical retrace - at the end of each frame the electron beam returns to the top-left corner. This is called vertical retrace.

### 3 Random Scan (Vector Scan)

In Random scan display the electron beam passes only to the part of screen where the picture is to be drawn rather than scanning from left to right bottom as in raster scan.

It is also called vector scan display / stroke-writing display or calligraphic display.

It is used to draw geometrical shapes such as points, lines, curves, polygons.

Images created using mathematical equations.

Used in line drawing applications.

Advantage of Random scan display is high resolution.

Random scan displays have higher resolution than raster scan display.

It produces smooth drawing.

Needs less memory.

Image quality is more in bitmapped.

On zooming picture quality remains same.

Disadvantage

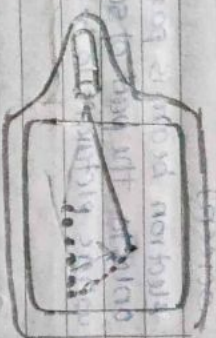
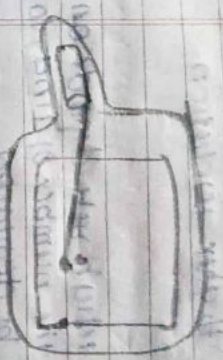
Only designed for graphical applications can not display realistic shaded scenes/images.

Not good for color image.

No shading.

Not good for complex pictures.

### Diagram



Random scan display

Random scan display

Random scan display

Random scan display

Random scan display

Random scan display

Random scan display

Random scan display

Random scan display

Random scan display

① Adv

# Difference b/w Raster Scan vs Random Scan display

| Feature                       | Raster Scan display                                                                                               | Random Scan display                                                                          |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| ① Resolution                  | Lower resolution                                                                                                  | Higher resolution                                                                            |
| ② Refresh rate                | Refresh rate is 60 to 80 frame per second (good film needed)                                                      | Refresh rate depends on the number of lines to be displayed (no need to refresh full screen) |
| ③ Electron beam               | Electron beam is painted line by line (row by row) from top left corner to the bottom right corner to draw image. | Electron beam is painted only to the part of screen where picture is to be drawn             |
| ④ Cost                        | Less expensive                                                                                                    | High cost (more expensive)                                                                   |
| ⑤ Image                       | It will produce for drawing image                                                                                 | Draw image using line.                                                                       |
| ⑥ Realism                     | Good for realistic image.                                                                                         | Not good for realistic image.                                                                |
| ⑦ Zooming                     | On zooming picture become blur                                                                                    | On zooming picture remain same.                                                              |
| ⑧ Line drawing                | Line drawing not very smooth                                                                                      | Line drawing very smooth.                                                                    |
| ⑨ Memory                      | Use frame buffer memory which is quick                                                                            | Use display list memory which is slow                                                        |
| ⑩ Complex screen easy to show | Complex screen easy to show                                                                                       | Complex screen difficult                                                                     |
| ⑪ Used in TV & monitors       | Used in TV & monitors                                                                                             | Used in CAD systems                                                                          |
| ⑫ Also called pre display     | Also called pre display                                                                                           | Vector display                                                                               |

## Difference between CRT and LCD (called Ray tube and Liquid crystal display)

| CRT                               | LCD                                                                                                                                                        |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ① CRT stands for Cathode Ray Tube | ① LCD stands for Liquid crystal display                                                                                                                    |
| ② CRT uses electron beam          | ② LCD uses liquid crystal                                                                                                                                  |
| ③ CRT is big & heavy              | ③ LCD is thin and light                                                                                                                                    |
| ④ takes more space                | ④ takes less space                                                                                                                                         |
| ⑤ more power consumption          | ⑤ less power consumption                                                                                                                                   |
| ⑥ produces more heat              | ⑥ produces less heat                                                                                                                                       |
| ⑦ CRT is less expensive           | ⑦ LCD more expensive                                                                                                                                       |
| ⑧ image formed by phosphor screen | ⑧ image formed by liquid crystal                                                                                                                           |
| ⑨ poor & low resolution           | ⑨ higher resolution.                                                                                                                                       |
| ⑩ Not portable device             | ⑩ easy to carry                                                                                                                                            |
| ⑪ old technology                  | ⑪ new technology.                                                                                                                                          |
| ⑫ used in old monitors, TV        | ⑫ used in modern monitors, laptops.                                                                                                                        |
| ⑬ Color CRT monitor               | ⑬ Color CRT monitor is a display device that produce colour image using electron beams & phosphor coated screen as it uses electron beam (e <sup>-</sup> ) |
| ⑭ Beam penetration method         | ⑭ Beam penetration method                                                                                                                                  |
| ⑮ Shadow mask method              | ⑮ Shadow mask method.                                                                                                                                      |

① Beam Penetration Method - (Random scan)

② Beam used in color CRT monitor, to produce color image.

③ Use three layers of phosphor, they are red phosphor and green phosphor.

④ Used in Random scan display.

⑤ Produce limited number of colors.

⑥ When the electron beam strikes outer phosphor layer, one color is produced and when it strikes the inner layer, another color is produced.

⑦ By changing the speed of electron diff. colors like red, green, orange & yellow can be produced.

⑧ The construction of beam penetration method is simple compared to shadow mask method.

⑨ It is cheaper.

⑩ Mostly used in line drawing applications.

⑪ Not for natural / realistic image (just like random scan).

⑫ Bcz of limited color, this method not used in modern computer monitor.

⑬ Disadvantage.

⑭ Only few colors can be produced.

⑮ Poor quality of picture (image) as blurry.

⑯ Not suitable for realistic image.

⑰ Not used in modern display device.

② Shadow Mask method. (Random scan)

③ It produces variety / wide range of colors as compared to beam-penetration method.

④ It is generally used in color beam display.

⑤ In this technique CRT has 3 phosphor (red dot at each pixel position).

⑥ One dot for red, one for green, one for blue (RGB). This is commonly known as 'Dot Triangle'.

⑦ A metal plate called shadow mask is placed in front of the screen.

⑧ The shadow mask has small holes which allow electron beams to hit the correct color spot.

⑨ When electron beam (carrying many different colors) can be produced on screen.

⑩ This method gives better picture quality & more realistic image compared to beam penetration method.

⑪ Can produce better millions of colors so it is suitable for graphics & video display.

⑫ More expensive & uses more power.

⑬ Used in TV & computer monitors.

⑭ Disadvantage / Draw.

⑮ High cost.

⑯ Complex structure.

⑰ Uses more electricity.

⑱ Produces more heat.

⑲ CRT becomes heavy.

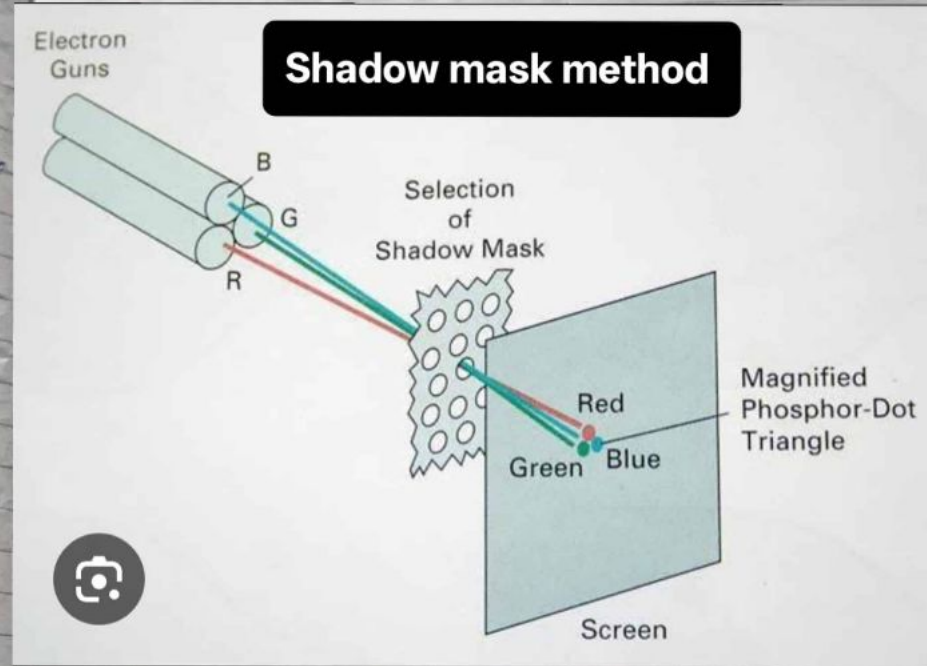
## Beam Penetration

- ① Two layers of phosphor dots (RGB) called Dot TRIANGLE
- ② Used in Random scan display
- ③ colors produced are red & green
- ④ Produce limited no of colors
- ⑤ construction is simple
- ⑥ low cost
- ⑦ low picture quality
- ⑧ so it is not used in modern display device
- ⑨ less power consumption
- ⑩ produces less heat
- ⑪ used for line drawing application

## Shadow Mask

- ① Two 3 phosphor color dots (RGB) called Dot TRIANGLE
- ② Used in Raster scan display
- ③ colors produced are red, green, blue
- ④ produce many colors
- ⑤ construction is complex
- ⑥ cost is high
- ⑦ High picture quality
- ⑧ used in most CRT monitor
- ⑨ more power consumption
- ⑩ produces more heat
- ⑪ used for TV & monitors

## Diagram - Shadow Mask Method.



## ⑤ Flat Panel Monitors

The type of video display device that have reduced volume, weight and power requirements compared to CRT.

Flat panel display is thin & light weight display used in TV, laptop & mobile.

It does not use big tube like CRT. It is flat in shape consume less power.

As flat panel display is thinner than CRT, we can hang them on wall or wear them on our necks.

Due can even work on some flat panel display so in future they maybe used like pocket notebooks.

Current use of flat panel display

- ① TV monitors
- ② Laptop
- ③ Smart watch
- ④ Calculators
- ⑤ Mobile phone

Two categories types of flat panel monitors displays

- ① emissive display → Plasma Panel
- ② Non-emissive display → Liquid Crystal

① Emissive display

- Emissive display is a display in which screen itself produce light to show image it does not need external light
- Emissive display convert electrical energy into light

Ex:- Plasma Panel, LED (light emitting diode), flat CRT

→ ① Plasma Panel / discharge display

①. This is also called as discharge display.

② Plasma display is a flat panel display that produce its own light for showing image.

③ Screen contains many small cells filled with gas like neon & xenon.

④ When electricity pulse, gas turns into plasma and emit light.

⑤ Each cell acts like a tiny bulb to create picture.

⑥ Plasma display gives very bright picture with good color quality.

⑦ This makes phosphor glow & forms images on screen

⑧ Viewing angle is wide, so picture can be seen clearly from all side

⑨ It is mostly used in large TVs & display board.

⑩ consume more power & produces more heat.

⑪ cost is high, heavy in weight and life is less than LCD.

⑫ Not commonly used in small device.

### Disadvantages

① High power consumption

② More heat generation

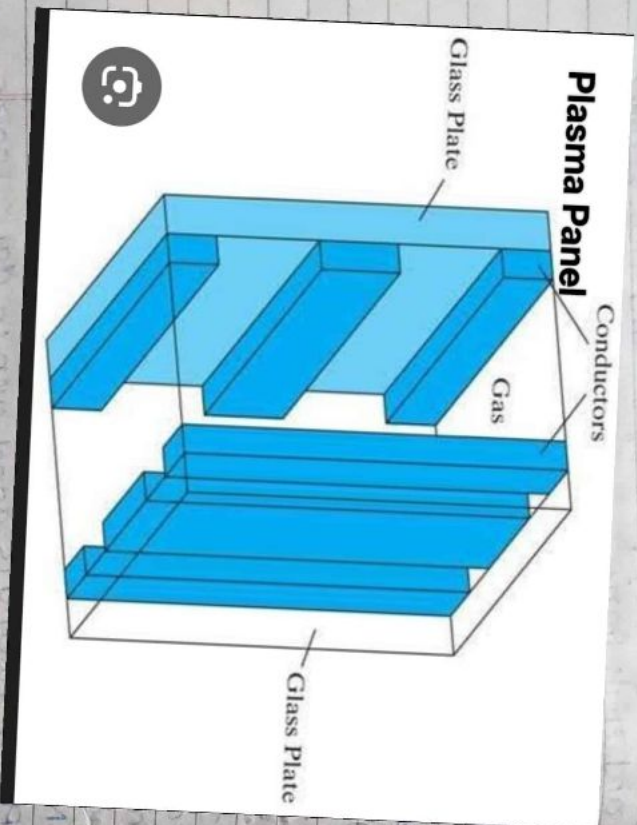
③ Expensive

④ Heavy weight compared to LCD.

⑤ Not suitable for small device.

Plasma panel display used with neon & argon gas. It produces its own light. It is called as discharge display.

## Diagram for plasma panel.



→ (2)

### Non-emissive Display,

- Non-emissive display does not produce its own light. It uses external light source to show image.
- Non-emissive use optical effect to convert sunlight from some other source into graphics pattern.
- ex: Liquid crystal display (LCD)

#### ① Liquid crystal display (LCD)

- LCD uses liquid crystal to form the image.
- Liquid crystal display is non-emissive display because it does not produce its own light.
- LCD uses an external light source called backlight to show image.
- Liquid crystals are placed b/w a glass plate.
- Liquid crystal can change position when electricity passes.
- The crystal control the light coming from backlight.
- By controlling light image are formed on screen.

#### Advantage

- LCD screen is very thin & light.
- LCD is light weight & easy to carry.
- LCD produces clear & sharp image.
- consumes less power & produce less heat.
- Used in laptop, mobility & monitors, calculator, digital watch, medical instruments.
- LCD does not produce radiation, so it is safe.
- Life of LCD is long.
- Cost is less compared to plasma.

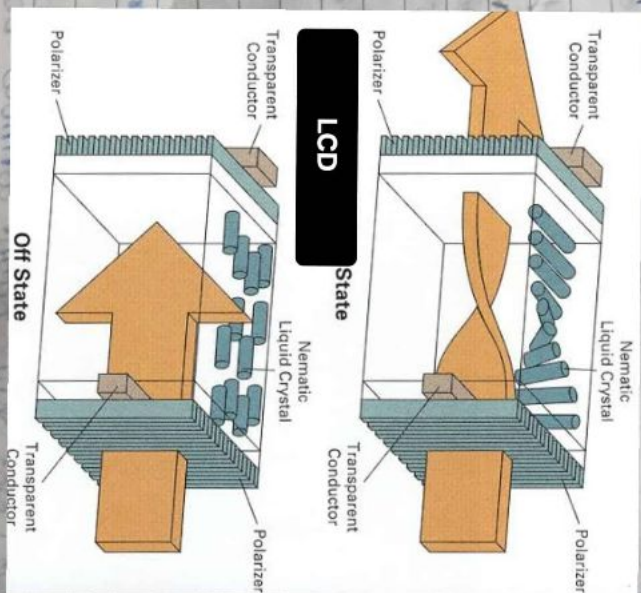
#### Disadvantage

- ① Viewing angle less than plasma.
- ② Response time slower than plasma.
- ③ Needs backlight to work.
- ④ Screen can break easily.
- ⑤ Not very bright without backlight.

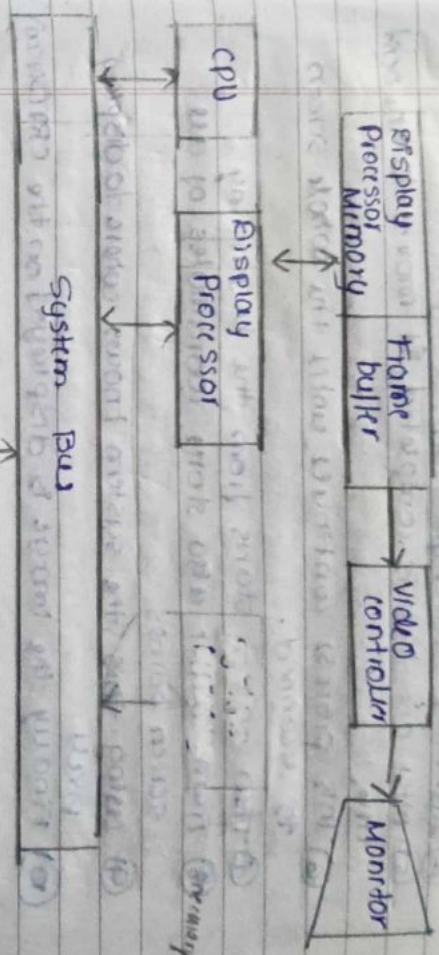
Working of LCD

1. The screen is covered by liquid crystals & backlight.
2. The LCD does not produce its own light, so it uses a light source behind the screen called backlight.
3. The screen has a glass plate and liquid crystal are placed between them.
4. When electric current passes through the liquid crystal, it changes their position.
5. These crystals control the amount of light passing through the screen.
6. When light passes through the crystals, it forms different colors & images.
7. Each small part of screen called a pixel and pixels together form the picture/image.
8. A color filter gives red, green, blue colors.
9. So in this way LCD shows clear images.
10. In power display image.

Diagram



## Architecture of Raster-Scan System



Definition: Raster scan system display image row by row on the screen.

- Raster graphics systems use different processing unit.
- Along with CPU, a special processor called video controller is used.
- The video controller reads data from frame buffer and shows it on screen.

The architecture consists of 3 main parts:

1. Frame buffer (store data)
  2. Video controller (display data)
  3. Display processor (process data)
- How it works: 1. The image to be displayed is stored in frame buffer memory. 2. The video controller reads the data from frame buffer and sends it to the display processor. 3. The display processor processes the data and sends it to the video controller. 4. The video controller sends the data to the monitor, which displays it on the screen.

NAME: \_\_\_\_\_  
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## Working of Raster Scan System

- ① The CPU gives graphics commands like draw line, circle to display processor.
- ② The graphics commands are in binary form 0 and 1 which is produced by display processor
- ③ Display processor converts graphics into pixel data to form image → called as scan conversion.
- ④ The result stored in Frame Buffer
- ⑤ The pixel data read by the video controller.
- ⑥ It send ~~pixel data~~ signals to the screen in proper orders  
It sends data to the monitor
- ⑦ It ~~scan~~ scan the screen line by line.  
~~from~~
- ⑧ After one line it moves to next line.  
(top to bottom)
- ⑨ The process continues until entire screen is ~~displayed~~ scanned
- ⑩ ~~to~~ After full screen it returns to top  
Vertical retrace
- ⑪ The screen refreshed continuously
- ⑫ Finally, the image is displayed on the screen

Three components / Parts <sup>①</sup> store image as pixel

### ① Frame Buffer / Refresh Buffer (Store)

- ① It is a part of system memory
- ② store color intensity of each pixel; color intensity
- ③ Acts like a picture storage of the screen
- ④ Picture definition is stored in a memory area called Frame Buffer / Refresh Buffer
- ⑤ It holds the complete image to be displayed
- ⑥ The fixed part of system memory is used as frame buffer

### ② Video Controller → (display)

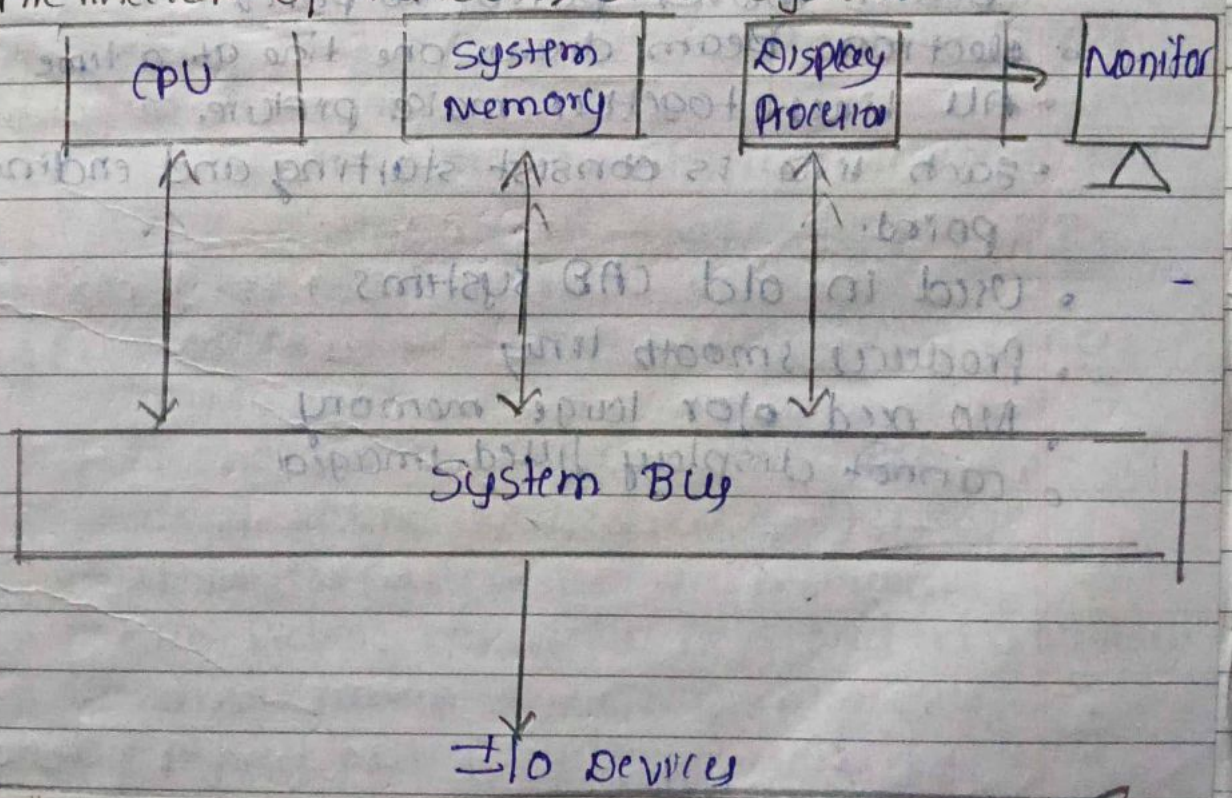
- ① video controller controls the display of image
- ② Read pixel data from frame buffer.
- ③ Send signals to the screen in proper order  
↓ It sends data to the monitor
- ④ Scan → It scans the screen line by line  
It displays image line by line
- ⑤ Refresh → It refreshes the screen continuously
- ⑥ Control → Control timing & speed,  
display 1 pixel at one time

③ Display Processor. <sup>which</sup> Display processor is a device which converts graphics into pixels to form image.

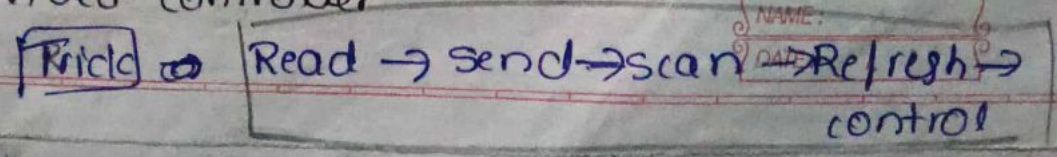
- It helps to reduce the work of CPU
- It performs operations like
  - line drawing
  - shape drawing
  - filling area.
  - displaying colors.
- Display processor is used to convert graphics data into image form
- It converts program instructions into pixel namely this process is called Scan conversion
- It store result of frame buffer.
- It is also used with input devices like Mouse Joystick

④ Scan conversion → process of converting graphics into pixel.  
 • It is done by display processor.

### Architecture of Random Scan System



### ② Video Controller



Memory I/O  
 (Read → Send — Scan — Refresh — control)

## Random Scan System

main. Random Scan System draws images using lines (vectors)

- It does not scan the full screen like raster

3 main components

- ① Display processor (process data)
- ② Refresh Display File (Store) (like frame buffer)
- ③ Vector generator (Display) (like video controller)
- ④ CRT Screen

### ① Display Processor (same point)

- Display processor converts graphics info

line drawing instruction (not pixels like raster)

- CPU gives commands to display processor.

- It reduce the work of CPU

- It performs operations like
  - line drawing
  - shape drawing

- It is used with input device like mouse

- Joystick

- The result stored in 'Refresh Display File'

### ② Refresh Display File (instead of frame buffer)

- ① It stores line drawing instruction (lines, curves) (not pixels)

- ② It stores lines & curves of image

- only holds required part of image

- It is a part of system memory

- The Picture definition is stored in a memory area called Refresh Display file

- A fixed part of system memory is used as Refresh display file

- ③ Each line & curves has color, intensity

## Vector Generator

1. It controls the display of image.
2. Read  $\rightarrow$  send  $\rightarrow$  scan  $\rightarrow$  Refresh  $\rightarrow$  control.

Vector generator Read line drawing instruction from Refresh display File.

It sends signals to the screen in proper order.

It sends data to the monitor.

Scan  $\rightarrow$  It scan only required part of image not full screen.

Refresh  $\rightarrow$  It refreshes the image continuously.

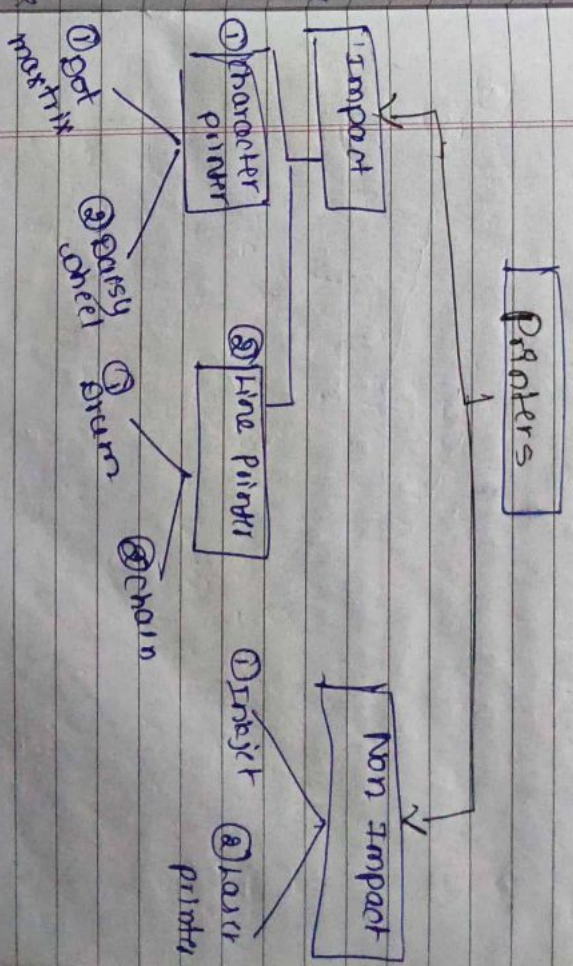
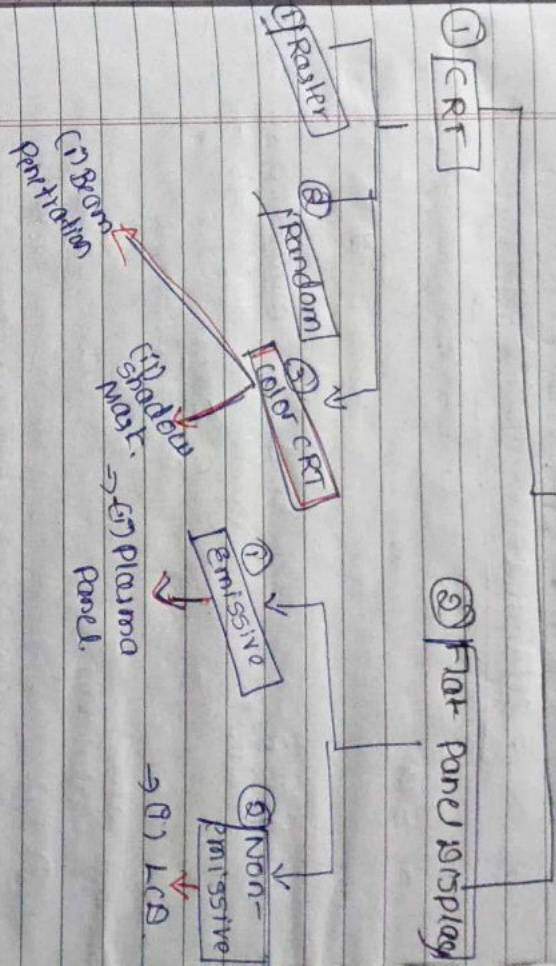
Control  $\rightarrow$  It controls the direction of electron beam.

## Working

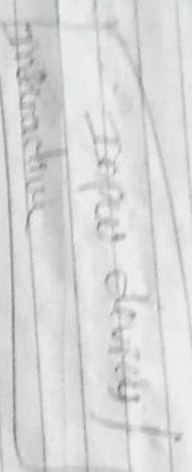
1. CPU gives graphics commands (line draw line, circle) to display processor.
2. The commands are in binary form 0 & 2 which is processed by display processor.
3. Display processor converts graphics into line drawing instruction.
4. The result stored in 'Refresh Display File'.
5. The instructions read by Vector generator.
6. It sends the signals to the screen in proper order.
7. The beam is used to draw lines on screen.
8. It draws only required part of the image then process called Random scan.
9. After drawing one line, it moves to next line.
10. It refreshes image continuously.
11. The Vector generator controls direction of electron beam.
12. Finally the image is displayed on screen.

1. Hard copy device  $\rightarrow$  Printers (only)
2. Input Device (only)

## Video Display Devices



3. Computerized Viewing Device  
is driven that is used to view 3D image



## Output Primitive

define own computer graphics, output primitive are the basic building blocks used to draw pictures.

• They help to create simple to complex images, shape, sizes.

• Output primitive are the basic shapes used in computer graphics to make images.

• They help the computer to draw pictures on the screen.

These include:

- Line
- Circle
- Polygon
- Point

• Simple shape or made by combining these.

• Each primitive needs input data such as coordinate & other details.

• This data tells the computer where to draw.

Points & Lines

• Points & Lines are the simplest shapes in computer graphics.

• All other shapes like polygon, circle are made by using them.

## POINTS



• Point is most basic graphical primitive. It represents position in a coordinate space.

• It has no size (no length or width).

• In a 2D space, point is defined by pair of coordinates.

• In a 3D space, point is defined by 3 coordinates.

• In a 2D space, point is defined by (x, y).

## ① Characteristics of points

① Location:- A point shows a fixed position on the screen

- Its location given by coordinates.

in 2D  $\rightarrow (x, y)$

in 3D  $\rightarrow (x, y, z)$

changing values of  $x, y, z$  will change position of point

② Size.

- A point has no actual size @ no length, no width, height or no dimensions

- On screen it looks like small dot (pixel)

- Size depends on display device resolution

- So point may look bigger or smaller

depending on screen size.

## ② Rendering Points

Rendering points means displaying a point on the screen.

① Pixel  $\rightarrow$  smallest dot on screen.

- In raster graphics, a point is displayed as 1 pixel (point is rendered as single pixel) on the screen.

- Each pixel has fixed position on screen

- Pixels have color and brightness

- Many pixels together form image.

② Point size  $\rightarrow$  how big dot appears

- Point is displayed as a single pixel on screen.

- Sometimes the point size can be increased

- Bigger point size means more pixels are used.

- Point size depends on system or graphics settings.

## II Line

- ① characteristic
- ② Rendering line
- ③ app'n.

- A line is a fundamental graphical primitive.
- ~~It~~ connects two <sup>distinct</sup> points.
- A line has length but no width.
- It is used to draw shapes and figures.
- It is made of many small points (collection of points over a path).

### ① Characteristics of Line

- ① Endpoints ② endpoints shows start & end of the line.

- ① A line is defined by 2 endpoints.

In 2D  $\rightarrow (x_1, y_1)$  and  $(x_2, y_2)$

In 3D  $\rightarrow (x_1, y_1, z_1)$  and  $(x_2, y_2, z_2)$

### ② Length and Direction

- Length is the distance between 2 endpoints.
- Direction shows which way the line goes.
- Longer distance  $\rightarrow$  long line.
- Shorter distance  $\rightarrow$  short line.
- A line can be horizontal, vertical, diagonal.

### ③ Thickness

- A line is thin in Theory.
- But on screen, it has some thickness.
- Thickness can be changed.
- It can be increased or decreased.
- Thin line  $\rightarrow$  1 pixel width.
- Thick line  $\rightarrow$  more pixel width.

### ② Rendering Line

- Rendering line means displaying line on a screen.

### ① Line Drawing Algorithms

- These are the methods/techniques <sup>steps</sup> to draw a line on the screen.
- There are 2 common algorithms:
  - (i) DDA  $\rightarrow$  Digital Differential Analyzer
  - (ii) Bresenham's Line Algorithm.
- These helps to make line smooth & accurate.
- They use pixels to form a straight line.
- They decide which pixels to turn on.

### ② Thickness and Style

- Lines can have different thickness.
  - Thin line  $\rightarrow$  1 pixel
  - Thick line  $\rightarrow$  more pixel.
- Lines can have different styles.
  - $\rightarrow$  Solid
  - $\rightarrow$  Dashed
  - $\rightarrow$  Dotted

- we can set different colors.
- Used to make line clear & attractive.

### ③ Anti-Aliasing

- <sup>zig-zag</sup> It is used to remove (jagged) edges (stair like).
- ~~used~~ It is used to make lines smooth by reducing rough edges.
- It is used when lines look rough.
- Improves overall picture quality.
- (Anti-Aliasing means making rough lines smooths)

### 3 Applications of Line

#### 1 Edges and Borders

- Lines are used to show edges and borders of shapes.
- They form boundaries/borders of the object.
- Helps to separate one object from other.
- Used to separate inside & outside area.
- Used to form tables, diagrams.
- Used in graphics & design.

#### 2 Graphs and charts

- Lines are used to draw graphs.
- Show increase & decrease in data.
- Used in graphs & bar charts.
- Used in reports, projects and presentation.
- Helps to understand data quickly.

#### 3 Design and Modelling

- Lines are used to create shapes & design.
- Used in computer-aided design.
- Used to make 2D & 3D designs.
- Modelling means used to design real world objects such as to design buildings, machines, etc.

### Line Drawing Algorithm

- Line drawing algorithm are the methods/techniques or steps used to draw a line on the screen.
- This helps to make line smooth & accurate.
- They use pixel to form straight line.
- They decide which pixels to turn on (to draw).
- Used to draw straight line between 2 points.

#### Requirements of Line Drawing algorithm

- 1 Line should appear as straight line.
- 2 Line should look straight.
- 3 It should be start & end accurately.

#### Different Types of Line

- 1 Horizontal line (Left-Right)
- 2 Vertical line (Up & down)
- 3 Diagonal line (45 degree)

#### Types of Line Drawing Algorithm

- 1 DDA → Digital Differential Analyzer.
- 2 Bresenham's line algorithm.

#### 1 DDA → Digital Differential Analyzer

- DDA is a simplest line drawing algorithm.
- DDA is a popular algorithm used to generate line, triangle & polygon.

#### 2 DDA is line drawing algorithm used in CG to generate line segment between 2 specified endpoints.

DDA is simple line drawing algorithm that draws a line by slowly increasing x and y values step by step from starting point to ending point by slowly changing (increasing) x & y values.

## Line Equation

Equation for a straight line  $\rightarrow$

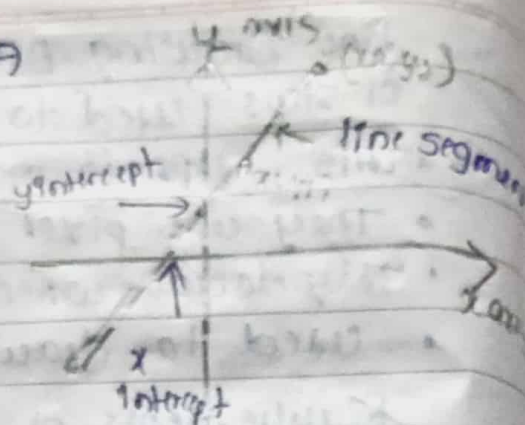
$$y = mx + c$$

$m \rightarrow$  slope

$c \rightarrow$  y-intercept

(where line segment/line

cuts y-axis)



Slope ( $m$ )

Given end points of line  $(x_1, y_1)$   $(x_2, y_2)$

$$m = \frac{(y_2 - y_1)}{(x_2 - x_1)} \Rightarrow m = \frac{dy}{dx} \Rightarrow$$

$$m = \frac{\Delta y}{\Delta x}$$

Relation between  $\Delta x$  and  $\Delta y$

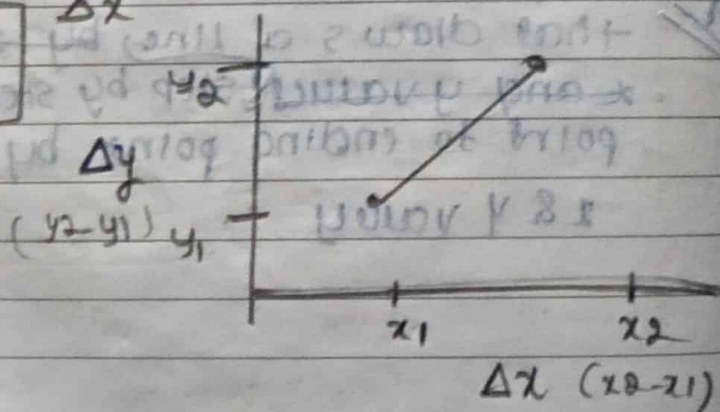
• If we know  $\Delta x \rightarrow$  we can find  $\Delta y$

$$m = \frac{\Delta y}{\Delta x}$$

$$\Delta y = m \cdot \Delta x$$

• If we know  $\Delta y \rightarrow$  we can find  $\Delta x$

$$\Delta x = \frac{\Delta y}{m}$$



stop condition

3 cases of Line (Based on slope  $m$ ) (note)

① case 1 : if  $m > 1$

(angle more than  $45^\circ$ )

$$x_{i+1} = x_i + 1$$

$$y_{i+1} = y_i + m$$

| $m > 1$                                                                                                                                                  | $m = 1$                                                                                                                                                  | $0 < m < 1$                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x_{i+1} = x_i + 1$                                                                                                                                      | $x_{i+1} = x_i + 1$                                                                                                                                      | $x_{i+1} = x_i + 1$                                                                                                                                      |
| $y_{i+1} = y_i + m$                                                                                                                                      | $y_{i+1} = y_i + 1$                                                                                                                                      | $y_{i+1} = y_i + m$                                                                                                                                      |
| <ul style="list-style-type: none"> <li>increase <math>x</math> by <math>1</math> step</li> <li>increase <math>y</math> by <math>m</math> step</li> </ul> | <ul style="list-style-type: none"> <li>increase <math>x</math> by <math>1</math> step</li> <li>increase <math>y</math> by <math>1</math> step</li> </ul> | <ul style="list-style-type: none"> <li>increase <math>x</math> by <math>1</math> step</li> <li>increase <math>y</math> by <math>m</math> step</li> </ul> |

② Initialize starting point

$$x = x_0$$

$$y = y_0$$

(get initial point of line as  $(x_0, y_0)$ )

③ Not the first point

(not no)  $(x, y)$  point

Procedure for DDA.

① Take input of the endpoints of the segment  
 Enter starting point  $\rightarrow (x_1, y_1)$  and endpoints  
 " ending point  $\rightarrow (x_2, y_2)$

② Calculate difference  $\rightarrow$  b/w the x and y  
 coordinates  $\Delta x$  and  $\Delta y$  respectively  
 $\Delta x = x_2 - x_1$   
 $\Delta y = y_2 - y_1$

③ Find m  
(Slope)  $\rightarrow$  calculate slope of line as  $m = \frac{\Delta y}{\Delta x}$

④ Find steps  
 if  $|\Delta x| > |\Delta y|$  steps =  $\Delta x$   
 else steps =  $\Delta y$

steps = maximum of  $|\Delta x|$  and  $|\Delta y|$

⑤ Find increment values calculate increment  
 $x \text{ increment} \rightarrow x_{inc} = \frac{\Delta x}{\text{steps}}$

$y \text{ increment} \rightarrow y_{inc} = \frac{\Delta y}{\text{steps}}$

⑥ Initialize starting point

set  $x = x_1$

$y = y_1$

(set initial point of line as  $(x_1, y_1)$ )

⑦ Plot the first point (Tably)

Draw point  $(x, y)$  (on tably)

plot round  $(x)$ , round  $(y)$

⑧ Generate next points

$x_{new} = x + x_{inc}$   
 $y_{new} = y + y_{inc}$   
 Plot each new point  $(x, y)$

⑨ Repeat

Continue until end point is reached.

⑩ Draw graph.

⑪ stop

Formula Algorithm  
 Read  $(x_1, y_1)$   $(x_2, y_2)$

①  $\Delta x = x_2 - x_1$

$\Delta y = y_2 - y_1$

② steps

if  $|\Delta x| > |\Delta y|$  then steps =  $|\Delta x|$   
 else steps =  $|\Delta y|$

steps =  $\max(|\Delta x|, |\Delta y|)$

③ increment

(i)  $x_{inc} = \frac{\Delta x}{\text{steps}}$

(ii)  $y_{inc} = \frac{\Delta y}{\text{steps}}$

④ Repeat all steps until end point is reached

addition along with long for

Problem ① BDA line drawing (5,4) (12,7)

- ① BDA line drawing (5,4) (12,7)  
 $x_1, y_1$   $x_2, y_2$

②  $\Delta x$  and  $\Delta y$

$$\Delta x = x_2 - x_1 = 12 - 5 = 7$$

$$\Delta y = y_2 - y_1 = 7 - 4 = 3$$

③ Slope (m)

$$m = \frac{\Delta y}{\Delta x} = \frac{3}{7} = 0.4$$

④ steps

$$if \text{abs}(\Delta x) > \text{abs}(\Delta y)$$

$$\text{steps} = \Delta x$$

$$\text{steps} = 7$$

⑤  $x_{inc}$  and  $y_{inc}$

$$x_{inc} = \frac{\Delta x}{\text{steps}} = \frac{7}{7} = 1$$

$$y_{inc} = \frac{\Delta y}{\text{steps}} = \frac{3}{7} = 0.4$$

⑥ ~~Plot~~ Initialize starting point

$$(5,4) \text{ as starting point}$$

$$x_1 = 5$$

$$y_1 = 4$$

⑦ Plot the first point on table

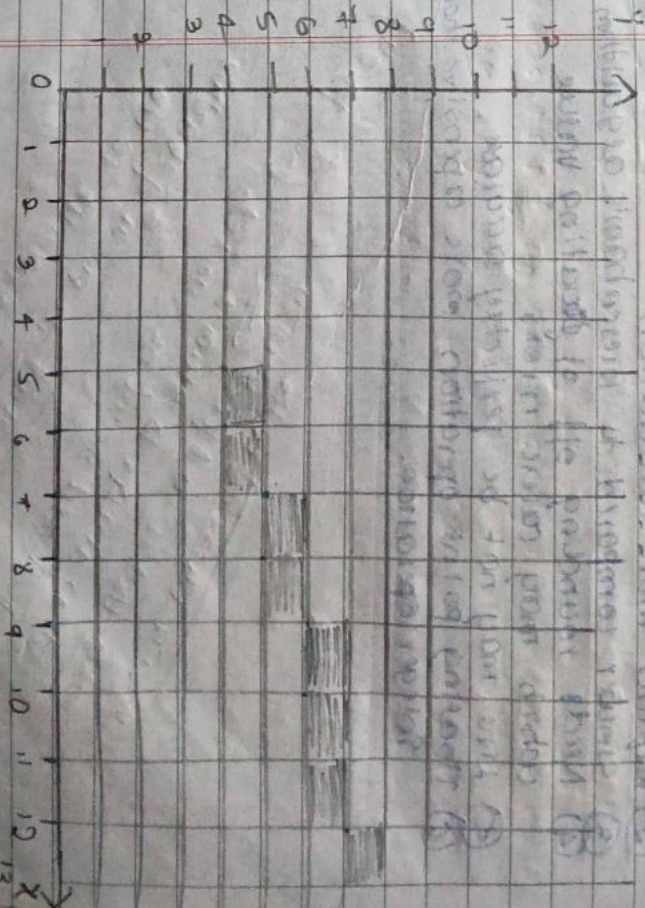
Table

| steps | x  | y   | round up |
|-------|----|-----|----------|
| 0     | 5  | 4   | 4        |
| 1     | 6  | 4.4 | 4        |
| 2     | 7  | 4.8 | 5        |
| 3     | 8  | 5.2 | 5        |
| 4     | 9  | 5.6 | 6        |
| 5     | 10 | 6   | 6        |
| 6     | 11 | 6.4 | 6        |
| 7     | 12 | 6.8 | 7        |

$$x_{inc} = 1 \quad y_{inc} = 0.4$$

$$x_1 = 5 \quad y_1 = 4$$

$$x_2 = 12 \quad y_2 = 7$$



## Advantage of DDA

1. The DDA is one of the simplest line drawing algorithm to use & implement.
2. It works on all types of lines.
3. It works efficiently for lines with a slope less than or equal to one.
4. Since it uses floating point values, we need to round off them for pixels.
5. DDA is incremental algorithm which means calculation are done step by step.

## Disadvantage of DDA

1. Use floating point values, which can be slow on some system.
2. Requires more calculation.
3. Slower compared to Bresenham's algorithm.
4. Needs rounding off of floating value which may cause errors.
5. Line may not be perfectly accurate.
6. Rounding point operation more expensive than integer operation.

Decision parameter is the value that is used for deciding whether the next pixel should be rounded up or down. It is calculated by adding the slope of the line to the current pixel value and then rounding it to the nearest integer.

## Bresenham's Line Generation Algorithm

Bresenham's line generation algorithm is a method used to draw a straight line by choosing the nearest pixel using only integer calculation. It is a fast method compared to DDA. It selects the best pixel to make the line look straight.

### Basic Idea

1. The algorithm draws a line step by step.
2. At each step, it chooses the nearest pixel to the actual line.
3. It can draw lines by increasing x or y by 1 unit.
4. It uses addition & subtraction to find the error.

Decision: The difference between the real line & chosen pixel is calculated. This difference is called 'decision parameter'.

Parameter / Decision variable (P) is  $P_0 = \Delta y - \Delta x$

Line equation: The algorithm starts with basic line equation  $y = mx + c$

1. It starts with line equation

2. Sampling: Sampling means converting a continuous line or shape into small pixels (points) on the screen.

3. Error calculation: The algorithm checks the difference between the pixel is chosen the real line.

4. Decision making: It is a process of deciding which pixel is closest to the real line. The algorithm chooses the pixel which is closest to the real line.

5. Efficient implementation method: only use simple addition & subtraction. So it is fast & easy to calculate.

6. Efficient implementation method

only use simple addition & subtraction.

# Procedure of Bresenham's Algorithm

1. Take input  
 Take starting point  $(x_0, y_0)$   
 Take ending point  $(x_1, y_1)$

2. Calculate difference dx and dy  
 $dx = x_1 - x_0$   
 $dy = y_1 - y_0$

3. Find m (slope)  
 $m = \frac{dy}{dx}$

check whether  $m < 1 \rightarrow$  use case 1  
 $m > 1 \rightarrow$  use case 2

or  
 not necessary to write this remainder

Case 1:  $m < 1$

1.  $x_1 = x_0 + 1$

$y_1 = y_0$

$P_1 = P_0 + 2dy$

2.  $x_1 = x_0 + 1$

$y_1 = y_0$

$P_1 = P_0 + 2dy - 2dx$

$x_1 = x_0 + 1$

$y_1 = y_0$

$P_1 = P_0 + 2dx - 2dy$

3. Initial Decision Parameter  
 $P_0 = 2dy - dx$

4. Plot first point on table  
 Plot first point, start from  $(x_0, y_0)$

5. Pixel selection

1.  $(P < 0)$  or  $(P > 0)$  Based on decision  
 choose next pixel horizontally or diagonally

2.  $(P < 0)$  choose next pixel horizontally

3. Update decision parameter  
 $P_1 = P_0 + 2dy$  if  $m < 1$   
 $P_1 = P_0 + 2dy - 2dx$  if  $m > 1$

4. Repeat  
 continue the steps until end point is reached.

5. Draw graph

### Algorithm formula.

① read  $(h'(x))$  and  $(e'h(x))$  pass

$$x^2b = 8qx - qx$$

②  $12 - 2x = xp$

$14 - 2h = hp$

③  $mp = \frac{dy}{dx}$

④ check  $m < 1$  or  $m \geq 1$

⑤ calculate  $P_0 = 2dy - dx$

↓  
decision parameter

if  $P_0 \leq 0$  then  
horizontal

if  $P_0 > 0$  then vertical

if  $m < 1$

$[1] (b < 0)$

$b' = b^0 + 8qx$

if  $m \geq 1$

$[0] (b < 0)$

$b' = b^0 + 8qx$

$[1] (b \geq 0)$

$b' = b^0 + 8qx - 8qx$

$[0] (b \geq 0)$

$b' = b^0 + 8qx$

Decision parameter is updated as per table.

Continue the steps until end point is reached.

### 4) Scaling

Scaling is a transformation which changes the size of the object

- It increase or decrease the size of an object
- Shape of the object remains same.
- Scaling performed by giving scaling factors  $S_x$  and  $S_y$

#### 3) Formula

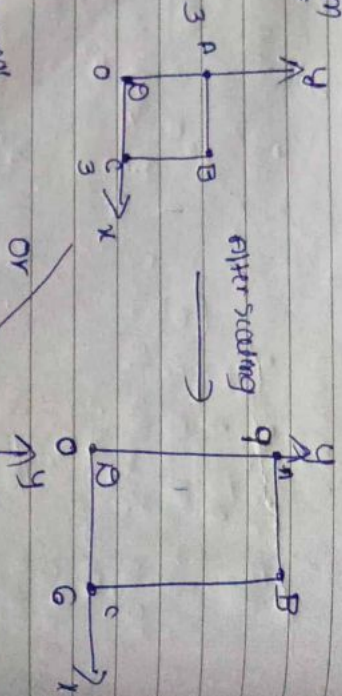
$$X' = S_x \cdot X$$

$$Y' = S_y \cdot Y$$

where  $(X', Y')$  new coordinate  $(X, Y)$  old coordinate  $S_x$  and  $S_y$  are scaling factors

3) Matrix rep<sup>n</sup> = 
$$\begin{bmatrix} X' \\ Y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$$

#### 2) Diagram



Applying  $x=4, y=4$  scaling factor

ex.  $P(2,3)$  scaled by  $S_x = 2$  becomes  $A(4,6)$

### 5) Shear Transformation

Shearing is a process of slanting the object in particular direction

- Change the shapes of the object
- Shearing can be performed along X axis or Y axis.

#### 3) Formula

2) X shear eq<sup>n</sup> 
$$X' = X + S_h \cdot Y$$

3) Y shear eq<sup>n</sup> 
$$Y' = Y + S_v \cdot X$$

1) Object is slanted along this axis

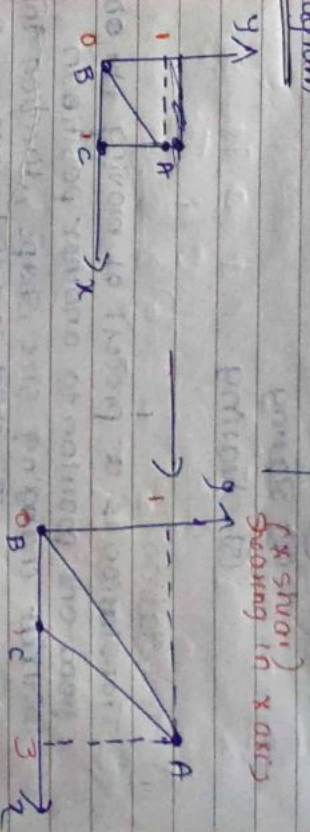
2) X axis is constant  
3) Y remains changing

#### 2) Matrix rep<sup>n</sup>

X shear eq<sup>n</sup> 
$$\begin{bmatrix} X' \\ Y' \end{bmatrix} = \begin{bmatrix} 1 & S_h \\ 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$$

Y shear eq<sup>n</sup> 
$$\begin{bmatrix} X' \\ Y' \end{bmatrix} = \begin{bmatrix} 1 & S_v \\ 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$$

#### 3) Diagram



Example  $P(2,3)$  with  $S_h = 1$  becomes  $(3,3)$

4) Application: - (4) Animation

## 2D Transformation

Transformation: Transformation is a process of changing the position, size, orientation of an object.

2D Transformation: are mathematical operations used in computer graphics to change position, size, orientation of an object. In 2D.

- used in 2D & 3D
- change object properties like position, size, shape.
- Represented using matrices

### Basic 2D Transformation

2D Transformations are mathematical operations used in CG to change position, size, orientation of an object in 2D.

Types of 2D Transformation

- 1) Translation
- 2) Reflection
- 3) Rotation
- 4) Scaling
- 5) Shearing

#### 1) Translation

Translation is a process of moving the object from one position to another position without changing size, shape, direction in 2D.

- The object is shifted along x axis only.

#### Formula

$$x_{\text{new}} = x_{\text{old}} + tx$$

only position change and size, shape remain same

$$y_{\text{new}} = y_{\text{old}} + ty$$

let old coordinates = (x<sub>old</sub>, y<sub>old</sub>)

$$y' = y + ty$$

Translation distance (tx, ty)  
New coordinates (x<sub>new</sub>, y<sub>new</sub>)

#### 1) Translation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x + tx \\ y + ty \end{bmatrix}$$

old coordinates (x, y)  
New coordinates (x', y')

#### matrix representation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} tx \\ ty \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} tx \\ ty \end{bmatrix}$$

#### 3) Rotation

$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

#### 4) Scaling

$$x' = x \cdot sx$$

$$y' = y \cdot sy$$

#### matrix

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} sx & 0 \\ 0 & sy \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

## ② Reflection

- Reflection about x axis

Formula

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ -y \end{bmatrix}$$

matrix

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

## ③ Shearing

$$x' = x + shx \cdot y$$

$$y' = y$$

matrix

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & shx \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

## ④ Translation

defn  
points

Formula + points

matrix representation

Diagram

## ⑤ Example

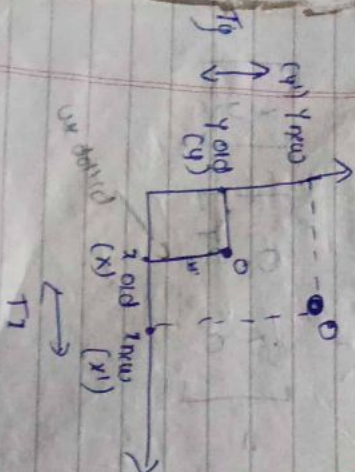
Point P(2,3)

translated by (4,2)

becomes (6,5)

## ⑥ Application

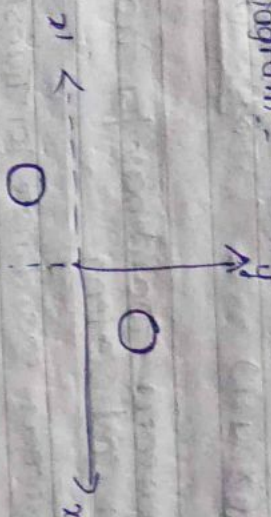
- used in moving objects in animation
- CAD



## ③ Reflection

- The size & shape of object remains same.
- Reflection can be performed on axes, parts or origin

## ④ Diagram



Example: 1 point (3,4) becomes (3,-4)

Appn: create mirror image

## ③ Rotation

- defn: Turning of an object clockwise or anticlockwise by a certain angle.
- Rotation change the orientation of object without changing its shape or size
- Rotation angle ( $\theta$ ) is used to rotate an object by specified angle

## ④ Formula

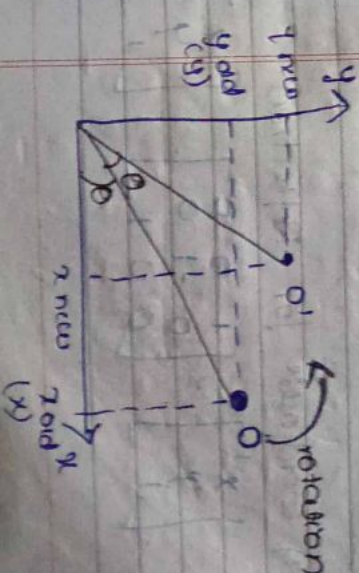
matrix representation

Diagram

## ⑤ Example

Example: 1 point (1,0) by  $90^\circ$  about the origin gives (0,1)

Appn: Animation, gaming, CAD image processing



# Homogeneous coordinate

1. A method of representing points by adding one extra coordinate i.e.  $(x, y)$

homogeneous  $(x, y, 1)$

## Homogeneous coordinate for translation, rotation and scaling

### 1 Homogeneous coordinate for Translation

1. def'n of Translation: Homogeneous translation is a transformation which moves object from one position to other position without changing size direction

2. formula -  $x' = x + tx$   
 $ty' = y + ty$

### 3 matrix representation:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} tx \\ ty \end{bmatrix}$$

### 4 homogeneous coordinate

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

### 5 diagram

### 1 Homogeneous coordinate for Rotation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

## Homogeneous coordinate convert all transformation into 3x3 matrix form.

- widely used in CG, animation & game.

### 2 Rotation

- 1. def'n: turning / rotating the object clockwise or anticlockwise by a certain angle.
- 2. formula:  $x' = x \cos \theta - y \sin \theta$   
 $y' = x \sin \theta + y \cos \theta$

### 3 matrix representation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

### 4 homogeneous

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

### 5 diagram

## Translation, Rotation, Scaling

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

### ③ Scaling

① defn Scaling is a transformation which changes the size of the object

formula =  $X' = X \cdot SX$   
 $Y' = Y \cdot SY$

matrix  $\begin{bmatrix} X' \\ Y' \end{bmatrix} = \begin{bmatrix} SX & 0 \\ 0 & SY \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$

### in homogeneous coordinate

$$\begin{bmatrix} X' \\ Y' \\ 1 \end{bmatrix} = \begin{bmatrix} SX & 0 & 0 \\ 0 & SY & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

### ⑤ Homogeneous coordinates for shearing

X shear  
 $X' = X + Shx \cdot y$   
 $y' = y$

matrix  $\begin{bmatrix} X' \\ Y' \end{bmatrix} = \begin{bmatrix} 1 & Shx \\ 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$

Homogr -  $\begin{bmatrix} X' \\ Y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & Shx & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$

Y shear  
 $y' = y + Shy \cdot x$   
 $x' = x$

matrix  $\begin{bmatrix} X' \\ Y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ Shy & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$

homogeneity  $\begin{bmatrix} X' \\ Y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ Shy & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$

### ③ Composite Transformation

① defn Composite transformation is combining two or more transformations such as translation, rotation, scaling and reflection is called composite transformation.

Ex: - 1 | an object is first translated and then rotated, this could be combined as a composite transformation

It is also called as concatenation of matrices. Matrix Representation for successive transformation scaling and rotation

### (i) Two successive rotation

① I & rotation angle is  $\theta_1$  and  $\theta_2$

② I an object is rotated by  $\theta_1$  and  $\theta_2$  then by total rotation  $\theta = \theta_1 + \theta_2$

$R(\theta_2) \times R(\theta_1) = R(\theta_1 + \theta_2)$

where  $R =$  Rotation matrix  
 First rotation =  $\theta_1$ ,  $\theta =$  rotation angle  
 and  $\theta_2 =$  Second rotation angle

②  $R_1 = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{bmatrix}$

Rotation matrix  $R_2 = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 \\ \sin \theta_2 & \cos \theta_2 \end{bmatrix}$

Composite rotation: ① and ② rotation is  $R_1(\theta_1)$  and  $R_2(\theta_2)$

## Composite rotation/scaling

$$R_2 R_1 \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Result} = R(\theta_2) R(\theta_1) = R(\theta_1 + \theta_2)$$

Conclusion: 2 successive rotation operations are additive.

(ii) two successive scaling transformation  
Suppose first scaling =  $S(S_{x1}, S_{y1})$

$$\text{Second scaling} = S(S_{x2}, S_{y2})$$

### Scaling matrices

$$S_1 = \begin{bmatrix} S_{x1} & 0 & 0 \\ 0 & S_{y1} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

① Scaling matrix  
② formula for scaling  
③ homogeneous coordinates  
④ 2D transformation  
⑤ diagram

$$S_2 = \begin{bmatrix} S_{x2} & 0 & 0 \\ 0 & S_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

### Composite scaling

$$S_2 S_1 = \begin{bmatrix} S_{x2} & 0 & 0 \\ 0 & S_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} S_{x1} & 0 & 0 \\ 0 & S_{y1} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Result

$$S_2 S_1 = \begin{bmatrix} S_{x1} S_{x2} & 0 & 0 \\ 0 & S_{y1} S_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

or

$$S(S_{x1}, S_{y1}) S(S_{x2}, S_{y2}) =$$

$$S(S_{x1} S_{x2}, S_{y1} S_{y2})$$

Conclusion: Two successive scaling operations are multiplicative.

(iii) two successive translation transformation

Suppose first translation =  $T(T_x, T_y)$   
and translation =  $T(T_{x2}, T_{y2})$

### Translation matrices

$$T_1 = \begin{bmatrix} 1 & 0 & T_{x1} \\ 0 & 1 & T_{y1} \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_2 = \begin{bmatrix} 1 & 0 & T_{x2} \\ 0 & 1 & T_{y2} \\ 0 & 0 & 1 \end{bmatrix}$$

### Composite translation

$$T_2 T_1 = \begin{bmatrix} 1 & 0 & T_{x2} \\ 0 & 1 & T_{y2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & T_{x1} \\ 0 & 1 & T_{y1} \\ 0 & 0 & 1 \end{bmatrix}$$

① translation matrix  
② formula  
③ matrix multiplication  
④ homogeneous coordinates  
⑤ successive operations  
⑥ diagram  
⑦ result

Result

$$T_3 T_1 = \begin{bmatrix} 1 & 0 & t_{x1} + t_{x2} \\ 0 & 1 & t_{y1} + t_{y2} \\ 0 & 0 & 1 \end{bmatrix}$$

Result:-  $T(t_2, t_{y2}) \cdot T(t_1, t_{y1}) =$

$$T(t_1 + t_2, t_{y1} + t_{y2})$$

conclusion:- Two successive translation operations are additive.

point

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

## Two Dimensional Viewing & Clipping

Window: The selected area of the world coordinate system.

Viewport: The area of the screen where the window is displayed

Clipping: The process of removing parts of an object outside the viewing area.

Types of clipping  $\rightarrow$  1. Point clipping  
2. Polygon clipping

Line Clipping  
The process of removing the unnecessary portions/parts of a line outside the viewing area.

Point Clipping  
The process of removing the unnecessary points that lie outside the viewing area and display only the points inside the window display area.

Polygon clipping  
The process of removing unnecessary parts of a polygon outside the viewing area.

Area Clipping  
The process of removing the unnecessary parts of an area outside viewing area.

Text Clipping: The process of removing text or part of a text which is outside viewing area & displaying only the text inside viewing area.

## Clipping Algorithm

- Clipping algorithm is a method used in CG to remove concave part of an object & to remove the viewing area and display the only the visible portion inside windows

### Two algorithms

- 1) Cohen-Sutherland line clipping algorithm
- 2) Sutherland-Hodgman Polygon Clipping Algorithm

### 1) Cohen Sutherland line clipping algorithm

Cohen Sutherland algorithm is a line clipping algorithm used to remove the portions of a line that lie outside the clipping window (viewing area)

- It divides the area around the clipping window is divided into 9 regions
- Each endpoint of the line is coded as 'region code'
- Using these codes the algorithm decides whether line is
  - Completely inside the window
  - It display it
  - Other wise it reject it
  - If it is partially inside window it clip & display visible part

0 0 0 1  $\rightarrow$  left  
 0 0 1 0  $\rightarrow$  R  
 0 1 0 1  $\rightarrow$  B  
 0 1 0 1  $\rightarrow$  B  
 1 0 0 0  $\rightarrow$  Top  
 (71,4) and (x2,y2)

Step 2: Find region code for both endpoints

Step 3: If both region code are 0000

accept the line because it is completely inside the clipping window

Step 4: If the logical AND of the 2 region

code's not 0000 (not zero)

code 1 AND code 2  $\neq$  0

then reject the line completely because it is completely outside the clipping window

Step 5: otherwise the line is partially inside & partially outside.

Step 6: Find the point where line meet the

window border (find where the line touch the clipping window)

Step 7: Replace the outside point with this border point. (at the outside part of the line)

Step 8: Recalculate the region code for the new endpoint

accepted or rejected

Step 9: Display the clipped (visible) portion of the line.

## Polygon Clipping Algorithm

### Algorithm

2) Sutherland-Hodgman algorithm is

- Sutherland-Hodgman algorithm is used for polygon clipping. method used to remove the unnecessary parts of the polygon that lie outside the clipping window and keep only the visible parts.

- It checks the polygon boundary by boundary.
- The portions outside the clipping window are removed.
- The remaining visible portion forms a new clipped polygon.

### Algorithm

Step 1: Input polygon vertices & clipping window.

Step 2: Check one side of the clipping window.

Step 3: If a polygon point is inside keep it.

Step 4: If a polygon point is outside, remove it.

Step 5: If an edge of polygon crosses the window boundary, find the crossing point.

Step 6: Add the crossing point to the new polygon.

Step 7: Repeat the same process for all sides of the clipping window.



Step 8: The remaining shape is the clipped polygon.

## Window to Viewport Transformation

Window to Viewport coordinate Transformation is the process of converting displaying the selected area of the world coordinate system (that is window) into a screen coordinate system.

Window: The selected area of the world coordinate system.

Viewport: The area of the screen where window is displayed.

### Main Idea

1. select window

2. select viewport

3. Convert window coordinates into viewport coordinates

4. Display the object in the viewport.

## Diagram

World coordinate

Window

Device coordinate

Viewport



xwmin xwmax

xwmin xwmax

## Algorithm

Step 1: Select the window in world coordinate

Step 2: Select the viewport on the display screen

Step 3: Calculate scaling factors in x and y direction

Step 4: Display the object in window coordinate into viewport coordinate

Step 5: Display the object in the viewport.

Window to Viewport Transformation

## Formula

$$X_v = X_{wmin} + (X_w - X_{wmin}) \cdot S_x$$

Y

The position of a point in the window should have same relative position in the view port, therefore

$$\text{Distance in the viewport} = \frac{\text{Distance in window}}{\text{viewport size}} = \frac{\text{window size}}{\text{viewport size}}$$

For x-coordinate

$$\frac{X_v - X_{vmin}}{X_{vmax} - X_{vmin}} = \frac{X_w - X_{wmin}}{X_{wmax} - X_{wmin}}$$

Cross multiplying

$$X_v - X_{vmin} = (X_w - X_{wmin}) \cdot \frac{X_{vmax} - X_{vmin}}{X_{wmax} - X_{wmin}}$$

Adding  $X_{vmin}$  both sides

$$X_v = X_{vmin} + (X_w - X_{wmin}) \cdot \frac{X_{vmax} - X_{vmin}}{X_{wmax} - X_{wmin}}$$

Similarly for y coordinate

$$Y_v = Y_{vmin} + (Y_w - Y_{wmin}) \cdot \frac{Y_{vmax} - Y_{vmin}}{Y_{wmax} - Y_{wmin}}$$

Final transformation eqn

$$X_v = X_{vmin} + (X_w - X_{wmin}) \cdot \frac{X_{vmax} - X_{vmin}}{X_{wmax} - X_{wmin}}$$

$$Y_v = Y_{vmin} + (Y_w - Y_{wmin}) \cdot \frac{Y_{vmax} - Y_{vmin}}{Y_{wmax} - Y_{wmin}}$$

# Polygon clipping

1) Defn - The process of removing unnecessary parts of a polygon that lie outside the clipping window (viewing area) and keeping only the visible part inside the window

2) explan - Polygon may be completely inside or partially inside the clipping window

- ① Polygon clipping removes the unnecessary portions outside windows
- ② remaining visible portion displayed
- ③ The Sutherland-Hodgman algorithm used for polygon clipping

## Not write about this algorithm

Before algorithm condition

3) Condition - (Case) for Sutherland-Hodgman Polygon

Case 1: Inside - Inside ( $C_i - I$ )

Keep the both vertices are inside the window  
→ keep the second vertex

Case 2 → Inside → Outside ( $I - O$ )

→ Add only the intersection point

Case 3 :- Outside → Inside ( $O - I$ )

→ Add intersection point & second vertex

Case 4 Outside → Outside ( $O - O$ )

→ Both vertices outside  
→ ignore and any point

4) After write algorithm - polygon clipping algorithm

## Application of clipping

- ① Computer graphics  
→ used to display only visible part of object
- ② CAD
- ③ Image processing
- ④ Presentation graphics
- ⑤ Computer art
- ⑥ Entertainment
- ⑦ Education training

same application of CG

Imp text & point clipping in chapter

## Text Clipping

defn

Method of Text clipping

① All-or-None string clipping

- Entire text is displayed if it is inside the window
- Entire text is rejected if any part is outside the window

② All-or-None character clipping

- Character inside the window are displayed
- Character outside are rejected

③ Text clipping

- Check parts of each character (string)
- Only the visible part inside window is shown

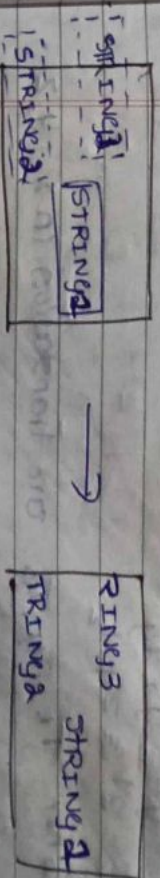
ex-1) Half of the letter 'H' is inside the window only half is displayed.

ex for ① All or None,

"HELLO" is partly outside window  
whole "HELLO" is not displayed

② All or None char

"HELLO" is partly outside only the character 'H' is shown



## Point Clipping

The rectangular window

defn

condition

①  $x_{min} \leq x \leq x_{max}$

②  $y_{min} \leq y \leq y_{max}$

If both conditions are satisfied the point is accepted otherwise it is rejected.

Steps ① Read the coordinates point (x, y)

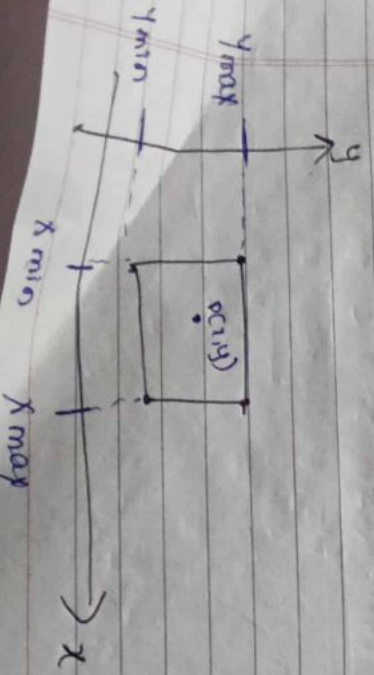
② check

$x_{min} \leq x \leq x_{max}$

$y_{min} \leq y \leq y_{max}$

③ If both conditions are true display point

④ otherwise reject it



# 3D Concepts

3D coordinate system  
coordinate system with x, y, z axes do 3D objects are called 3D

## 3D Transformation

3D Transformation is the process of changing the position, size, direction, orientation of a 3D object in space using mathematical operations

- It is used in CG to move, rotate, resize objects

Types of 3D Transformation - 3D Translation, 3D Rotation, 3D Scaling, 3D Shearing

## Basic 3D Transformation

### 1) Translation

3D translation moves the object from one position to another with 3D without changing its shape, size and direction

- only position change, size, direction remain same
- The object shift along x or y axis.

#### Formula

If a point is  $P(x, y, z)$  after translation it becomes  $P'(x', y', z')$

$$x' = x + t_x$$

$$y' = y + t_y$$

$$z' = z + t_z$$

where  $T_x, T_y, T_z$  are translation in x, y, z direction

Explain 3D transformation with respect to x, y, z axis and matrix representation  
Explain 3D Rotation

## Matrix form (homogeneous)

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Shape remains same, only position change

### 2) 3D Reflection

3D reflection creates mirror image of an object

Common reflection

• About XY-plane

$$(x, y, z) \rightarrow (x, y, -z)$$

Matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

#### Formula

• About XY-plane

$$\begin{matrix} x' = x \\ y' = y \\ z' = -z \end{matrix}$$

• About YZ-plane

$$(x, y, z) \rightarrow (-x, y, z)$$

• About XZ-plane

$$(x, y, z) \rightarrow (x, -y, z)$$

• About YZ-plane

$$\begin{matrix} x' = -x \\ y' = y \\ z' = z \end{matrix}$$

### 3D Rotation

about (x, y, z)

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

### 3D Rotation

about (x, y, z)

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

### 3) Rotation

Rotation is a turning an object clockwise or anticlockwise by certain angle in  $x, y, z$  direction

Rotation about x axis

y axis rotation

$$y' = y \cos \theta - z \sin \theta$$

$$z' = y \sin \theta + z \cos \theta \Rightarrow$$

$$x' = x$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

z axis rotation

$$z' = z \cos \theta - x \sin \theta$$

$$x' = z \sin \theta + x \cos \theta$$

$$y' = y$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

x axis rotation

$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

$$z' = z$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

### 4) Scaling

Scaling is the transformation which changes the size of the object in  $x, y$ , and  $z$  direction

$$x' = x \cdot s_x$$

$$y' = y \cdot s_y$$

$$z' = z \cdot s_z$$

where  $s_x, s_y, s_z$  are scaling factors

Matrix form

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Type

1) Uniform scaling : same factor in all direction

2) Non uniform scaling  $\rightarrow$  different factors

### 5) Shearing (skewing)

Shearing is the process of slanting object in a particular direction

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- It helps to see an objective on that way

## display technology

## Types of Projection

→ ① perspective projection

### Projection Cell Point

at a realistic view (human eye) 11/20/23

0 3 5 6 7 8

Diagram

① One-point perspective

⑨ Three - Point Perspective.

- only 1 vanishing point

- simple vit

Example 1. 1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100. 101. 102. 103. 104. 105. 106. 107. 108. 109. 110. 111. 112. 113. 114. 115. 116. 117. 118. 119. 120. 121. 122. 123. 124. 125. 126. 127. 128. 129. 130. 131. 132. 133. 134. 135. 136. 137. 138. 139. 140. 141. 142. 143. 144. 145. 146. 147. 148. 149. 150. 151. 152. 153. 154. 155. 156. 157. 158. 159. 160. 161. 162. 163. 164. 165. 166. 167. 168. 169. 170. 171. 172. 173. 174. 175. 176. 177. 178. 179. 180. 181. 182. 183. 184. 185. 186. 187. 188. 189. 190. 191. 192. 193. 194. 195. 196. 197. 198. 199. 200. 201. 202. 203. 204. 205. 206. 207. 208. 209. 210. 211. 212. 213. 214. 215. 216. 217. 218. 219. 220. 221. 222. 223. 224. 225. 226. 227. 228. 229. 230. 231. 232. 233. 234. 235. 236. 237. 238. 239. 240. 241. 242. 243. 244. 245. 246. 247. 248. 249. 250. 251. 252. 253. 254. 255. 256. 257. 258. 259. 260. 261. 262. 263. 264. 265. 266. 267. 268. 269. 270. 271. 272. 273. 274. 275. 276. 277. 278. 279. 280. 281. 282. 283. 284. 285. 286. 287. 288. 289. 290. 291. 292. 293. 294. 295. 296. 297. 298. 299. 300. 301. 302. 303. 304. 305. 306. 307. 308. 309. 310. 311. 312. 313. 314. 315. 316. 317. 318. 319. 320. 321. 322. 323. 324. 325. 326. 327. 328. 329. 330. 331. 332. 333. 334. 335. 336. 337. 338. 339. 340. 341. 342. 343. 344. 345. 346. 347. 348. 349. 350. 351. 352. 353. 354. 355. 356. 357. 358. 359. 360. 361. 362. 363. 364. 365. 366. 367. 368. 369. 370. 371. 372. 373. 374. 375. 376. 377. 378. 379. 380. 381. 382. 383. 384. 385. 386. 387. 388. 389. 390. 391. 392. 393. 394. 395. 396. 397. 398. 399. 400. 401. 402. 403. 404. 405. 406. 407. 408. 409. 410. 411. 412. 413. 414. 415. 416. 417. 418. 419. 420. 421. 422. 423. 424. 425. 426. 427. 428. 429. 430. 431. 432. 433. 434. 435. 436. 437. 438. 439. 440. 441. 442. 443. 444. 445. 446. 447. 448. 449. 450. 451. 452. 453. 454. 455. 456. 457. 458. 459. 460. 461. 462. 463. 464. 465. 466. 467. 468. 469. 470. 471. 472. 473. 474. 475. 476. 477. 478. 479. 480. 481. 482. 483. 484. 485. 486. 487. 488. 489. 490. 491. 492. 493. 494. 495. 496. 497. 498. 499. 500. 501. 502. 503. 504. 505. 506. 507. 508. 509. 510. 511. 512. 513. 514. 515. 516. 517. 518. 519. 520. 521. 522. 523. 524. 525. 526. 527. 528. 529. 530. 531. 532. 533. 534. 535. 536. 537. 538. 539. 540. 541. 542. 543. 544. 545. 546. 547. 548. 549. 550. 551. 552. 553. 554. 555. 556. 557. 558. 559. 560. 561. 562. 563. 564. 565. 566. 567. 568. 569. 570. 571. 572. 573. 574. 575. 576. 577. 578. 579. 580. 581. 582. 583. 584. 585. 586. 587. 588. 589. 590. 591. 592. 593. 594. 595. 596. 597. 598. 599. 600. 601. 602. 603. 604. 605. 606. 607. 608. 609. 610. 611. 612. 613. 614. 615. 616. 617. 618. 619. 620. 621. 622. 623. 624. 625. 626. 627. 628. 629. 630. 631. 632. 633. 634. 635. 636. 637. 638. 639. 640. 641. 642. 643. 644. 645. 646. 647. 648. 649. 650. 651. 652. 653. 654. 655. 656. 657. 658. 659. 660. 661. 662. 663. 664. 665. 666. 667. 668. 669. 670. 671. 672. 673. 674. 675. 676. 677. 678. 679. 680. 681. 682. 683. 684. 685. 686. 687. 688. 689. 690. 691. 692. 693. 694. 695. 696. 697. 698. 699. 700. 701. 702. 703. 704. 705. 706. 707. 708. 709. 710. 711. 712. 713. 714. 715. 716. 717. 718. 719. 720. 721. 722. 723. 724. 725. 726. 727. 728. 729. 730. 731. 732. 733. 734. 735. 736. 737. 738. 739. 740. 741. 742. 743. 744. 745. 746. 747. 748. 749. 750. 751. 752. 753. 754. 755. 756. 757. 758. 759. 760. 761. 762. 763. 764. 765. 766. 767. 768. 769. 770. 771. 772. 773. 774. 775. 776. 777. 778. 779. 780. 781. 782. 783. 784. 785. 786. 787. 788. 789. 790. 791. 792. 793. 794. 795. 796. 797. 798. 799. 800. 801. 802. 803. 804. 805. 806. 807. 808. 809. 810. 811. 812. 813. 814. 815. 816. 817. 818. 819. 820. 821. 822. 823. 824. 825. 826. 827. 828. 829. 830. 831. 832. 833. 834. 835. 836. 837. 838. 839. 84

- also a warning points

eg:- corner of building

- used 3 vanishing points

Q3) :- ~~used~~ for tall building, from ground

10

① very realistic

Coastal

89army.

Type of projection in which the objects do not <sup>change</sup>

- All projection lines are parallel to each other

Handwritten notes:

- of
- at
- the
- end
- of
- the
- road

# Types of Parallel Projection

- 1) Orthographic projection
- 2) Oblique projection

1) Orthographic projection ( $90^\circ$ ) to the projectors are perpendicular to the projection plane.

2) No angle in projection shows true shape, true size or parallel.

3) Used in engineering drawings

4) Used for front, top, side, view

5) Machine part drawing

2) Oblique Projection ( $45^\circ$ ) projection lines are not perpendicular to the projection plane.

The object is not view straight

The lines are tilted or slanted

5D the object slightly 3D.

Oblique projection are easy to draw

But this is less realistic than orthographic

Depth is shown

one side of the object is seen in its exact real shape & size

other side are slanted / tilted not exact

## Types of Oblique Projection

- 1) Cavalier Projection
- 2) Cabinet Projection
- 3) General Oblique projection

1) Cavalier projection

Depth is shown in full size (100%)

2) Cabinet projection

Depth is shown in half size (50%)

3) General oblique projection

Depth can be shown in any reduced scale

Ex: - custom 3D drawings.

## Advantages

- easy to draw
- very accurate
- useful in civil engineering

## Disadvantages

- 1) not realistic
- 2) no depth feeling

## Conclusion

Parallel projection used where accuracy is more important than realism

## Difference between Perspective & Parallel

| perspective Projection                                                                                           | parallel Projection                                                                                            |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| ① Perspective projection is a type of projection in which object look smaller when they are far from the viewer. | ① Parallel projection is a type of projection in which object do not change size whether they are near or far. |
| ② Projection lines meet at a vanishing point.                                                                    | ② Projection lines are parallel.                                                                               |
| ③ Distant objects appear smaller.                                                                                | ③ Object size remains same.                                                                                    |
| ④ Has vanishing point.                                                                                           | ④ No vanishing point.                                                                                          |
| ⑤ Does not give accurate measurement.                                                                            | ⑤ Gives accurate measurement.                                                                                  |
| ⑥ Used in realistic drawings.                                                                                    | ⑥ Used in engineering drawings.                                                                                |
| ⑦ More complex drawing.                                                                                          | ⑦ Easy to draw.                                                                                                |
| ⑧ Object look more realistic.                                                                                    | ⑧ Object look less realistic.                                                                                  |
| ⑨ Parallel lines may appear to meet.                                                                             | ⑨ Parallel lines may remain parallel.                                                                          |
| ⑩ Used in games, animation & photography.                                                                        | ⑩ Used in CAD engineering drawings.                                                                            |

## Octrees

Octrees An octree is a tree based data structure used in computer graphics to represent a 3D that divide 3D space into 8 equal partitions to represent object efficiently.

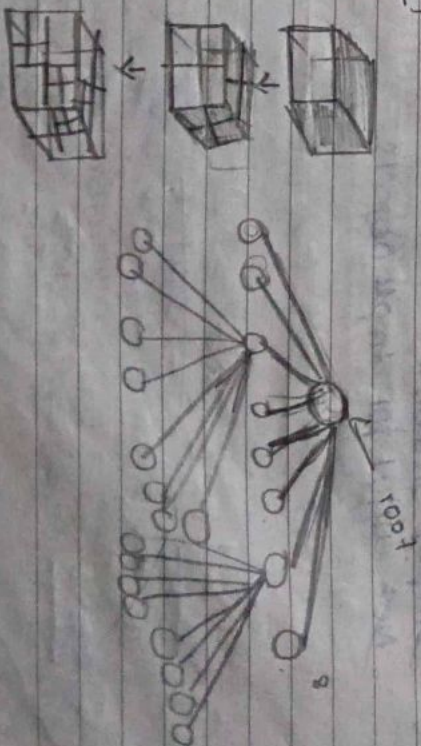
- The whole 3D space is taken as cube.
- This cube divided into 8 smaller cube.
- Each small cube can again be divided into 8 parts if needed.
- This continues until required detail achieved.

### Structure

- Each node has 8 children.
- Each child represents one part of 3D space.
- Top node = root.

- Travelling
- Efficient way to store 3D object.
  - Save memory by dividing only needed region.
  - Used for complex 3D scene.

### Diagram



### Type

#### ① Full Uniform Octree

- Every node is divided into exactly 8 equal parts.

#### ② Non Uniform (Adaptive)

- only required parts are subdivided
- save memory

### Application

- 3D computer graphic
- game development
- Robotics
- Image compression

### Advantage

- Reduce memory usage
- Fast searching in 3D

### Disadvantage

- complex structure
- Not useful for simple objects.

### 15M

#### Bezier Curve

explain with prob. and ex.  
 It is developed by Pierre Bézier engineer used in CAD graphic

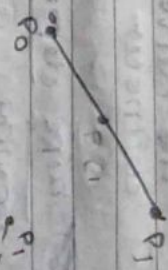
Defn: Bezier curve is a smooth curve used in computer graphics that is defined using control points.

- It is used to define the shapes like fonts, logos & animations

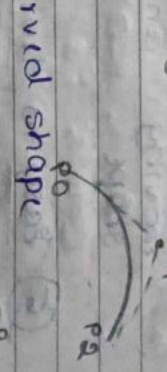
#### Types of Bezier curve

- ① Linear Bezier Curve
- ② Quadratic Bezier Curve
- ③ Cubic Bezier Curve

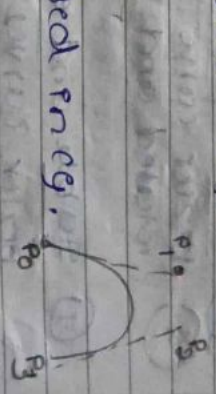
- ① Linear Bezier curve
  - uses 2 control points
  - produces straight line



- ② Quadratic Bezier Curve
  - uses 3 control points
  - produces a simple curved shape



- ③ Cubic Bezier Curve
  - uses 4 control points
  - mostly commonly used in eg.



#### Bezier Curve Formula

±] there are n+1 control points

$P_k = (x_k, y_k, z_k)$  then the Bezier curve is

$$P(u) = \sum_{k=0}^n P_k B_{k,n}(u) \quad 0 \leq u \leq 1$$

where:  $P_k$  = control points

- ②  $B_{k,n}(u)$  = Bernstein blending function
- ③  $u$  = Parameter

Properties of Bezier Curve

1. Passes through the first and last control points

2. Does not necessarily pass through intermediate control points

3. The curve has a smooth appearance without sharp corners

4. The curve is always convex

5. The curve always remains inside the polygon formed by control points

6. The curve has a smooth appearance without sharp corners

7. The curve has a smooth appearance without sharp corners

8. Shape depends on control points

9. Moving any control point changes the shape

10. Easy to transform Bezier curve can be easily translated, rotated and scaled.

11. Stable and easy to compute

12. Bezier curves are mathematically stable & simple to generate using Bernstein polynomials

13. Defined by control points

14. A Bezier curve is joined using set of control points.

15. Symmetry property  $\rightarrow$  If the control points are symmetric the Bezier curve is also symmetric

16. Degree property  $\rightarrow$  If there are  $n+1$  control points the degree of Bezier curve is  $n$

## Application

1. Animation

2. Font design

3. Logo design

4. CAD

5. Image editing software

## Circle

Circle is a set of all points in a plane that are equidistant from the fixed point called as Radius.

Property of circle -

1. All points are equidistant from the center every point on the circle is at the same distance from center. This distance is called as Radius ( $r$ )

2. Diameter is twice the radius

3. Circumference of a circle  $C = 2\pi r$

4. Area of a circle

5. Tangent property

6. The region enclosed by a circle is called Area

$$A = \pi r^2$$

7. Tangent property

8. The tangent is always perpendicular to the radius at the point of contact

6) Arc property  
Any part of the circumference is called an arc

two types  
① minor arc  
② major arc

7) Equal radii property  
All radii of the same circle are equal in length

8) Symmetric Property  
Circle is symmetric about

- x axis
- y axis

Circle Drawing Algorithm / Circle Drawing Techniques

Circle Drawing Algorithms are the methods or techniques used to draw circle on a computer screen accurately

Types of Circle Drawing Algorithm

- ① Bresenham's circle drawing algorithm
- ② midpoint circle drawing algorithm

9) 8 way symmetry property

Circle has 8 way symmetry  
If one point (x, y) is known seven other points can be obtained:

- ① (x, y)
- ② (-x, y)
- ③ (-x, -y)
- ④ (x, -y)
- ⑤ (x, y)
- ⑥ (-x, -y)
- ⑦ (-x, y)
- ⑧ (x, -y)

This property reduces computation in line drawing algorithms

① Bresenham's Circle Drawing Algorithm

Bresenham's circle drawing algorithm is a method used to draw a circle by selecting the nearest pixel position using a decision parameter.

It uses 8 way symmetry property of a circle

It only performs only integer calculation which makes it fast & efficient

Important feature: *uses only integer*

Circle equation  
 $x^2 + y^2 = r^2$

Decision parameter  $\rightarrow$  *decision parameter*  
Initial decision parameter  $\rightarrow$  *initial value used*

$P = 3 - 2r$  for decision making. It helps to determine which pixel to choose.

② Sampling  $\rightarrow$  The algorithm samples pixel positions to draw the remaining points of the circle and uses 8 way symmetry to draw the remaining points

③ Error calculation: The algorithm checks difference between actual circle & selected pixel position

④ Decision making: It is a process of deciding which pixels are closest to the circle.

If  $P < 0$ , choose East pixel.  
If  $P > 0$ , choose South-East pixel.

⑤ Efficient implementation: Implementing using only integer arithmetic to avoid multiplication and square root operation

## Algorithm Circle Drawing Algorithm

Step 1: Start

Step 2: Input radius  $r$  and center of the circle

Step 3: Initialize

$X_0 = 0$  (mid point)

$Y_0 = r$

$P = 3 - 2r$  (decision parameter)

Step 4: Plot initial points using symmetry

Step 5: Repeat until

$X_{i+1} > Y_{i+1}$

if  $(P < 0)$

$X_{i+1} = X_i + 1$

$Y_{i+1} = Y_i$

$P = P_0 + 4X_i + 6$

else

$X_{i+1} = X_i + 1$

$Y_{i+1} = Y_i - 1$

$P = P_0 + 4(X_i - Y_i) + 10$

Plot symmetric points

Step 6: Calculate and plot the coordinates for the remaining  $Q_1, Q_2, Q_3, Q_4$  (remaining 7 symmetric points)

Step 7: Stop.

explan

① The algorithm starts from top point of the circle  $(0, r)$

② A decision parameter  $P$  determines the next pixel position

③ if  $P < 0$ , only  $x$  is increased

④ if  $P > 0$ ,  $x$  is increased  $y$  is decreased

⑤ using 8 way symmetry only one eighth of the circle is calculated and remaining points are obtained by reflection

⑥ This makes the algorithm fast.

Mid point Circle Drawing Algorithm  
Mid point circle drawing algorithm is used to draw circles by selecting pixels based on using midpoint decision parameter & symmetry

Algorithm  $\rightarrow$  next page

## Mid point Circle Drawing Algorithm

Step 1: Start

Step 2: Input radius  $r$  and center of the circle.

Step 3: Initialize

$$x_0 = 0$$

$$y_0 = r$$

$$p = 1 - r$$

( $p$  is an integer)

or

$$p = \frac{5 - r}{4}$$

Step 4: Plot initial points using symmetry

Step 5: Repeat until  $x_{i+1} > y_{i+1}$

$$x_{i+1} > y_{i+1}$$

if ( $p < 0$ )

$$x_{i+1} = x_i + 1, y_{i+1} = y_i$$

$$y_{i+1} = y_i$$

$$p_1 = p_0 + 2x_{i+1} + 3$$

else

$$x_{i+1} = x_i + 1$$

$$y_{i+1} = y_i - 1$$

$$p_1 = p_0 + 2x_{i+1} - 2y_i + 5$$

Step 6: Plot the symmetry points

$Q_1, Q_2, Q_3, Q_4$  and the

remaining symmetric points

Step 7: Stop.

## Explanation

① The algorithm begins at (0,1)

② A midpoint algorithm checks the middle point plus 8 possible pixels and selects the pixel that is nearest to the circle boundary.

③ If the midpoint lies inside the circle, only  $x$  is increased

④ If the midpoint lies outside the circle,  $x$  is increased &  $y$  is decremented

⑤ The 8 way symmetry property is used to generate the complete circle.

3d display

Technique

change

(a) x and y are not in pixels

homogeneous matrix repn

1 Translation

formula

$$X' = X + Tx$$

$$Y' = Y + Ty$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} Tx \\ Ty \end{bmatrix}$$

2 Reflection

$$X' = X$$

$$Y' = -Y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

3 Rotation

$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

4 Scaling

$$X' = X \cdot Sx$$

$$Y' = Y \cdot Sy$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} Sx & 0 \\ 0 & Sy \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

5 Shearing

shearing transformation

x shear

$$X' = X + Shx \cdot y$$

$$Y' = y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & Shx \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

y shear

$$X' = x$$

$$Y' = y + Shy \cdot x$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 & Shy \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

homogeneous matrix translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & Tx \\ 0 & 1 & Ty \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Matrix Repn for 2 successive transformations

$$T_2 T_1 = \begin{bmatrix} 1 & 0 & Tx_1 + Tx_2 \\ 0 & 1 & Ty_1 + Ty_2 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Scaling

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} Sx & 0 & 0 \\ 0 & Sy & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

3D translation

1 Translation

$$X' = X + Tx$$

$$Y' = Y + Ty$$

$$Z' = Z + Tz$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & Tx \\ 0 & 1 & 0 & Ty \\ 0 & 0 & 1 & Tz \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

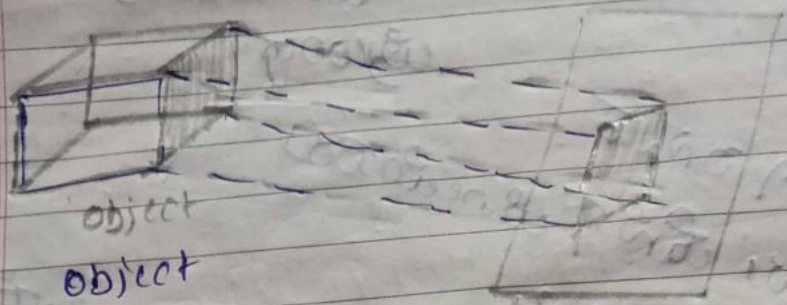
2 Rotation - Z axis

$$X' = X \cos \theta - y \sin \theta$$

$$Y' = X \sin \theta + y \cos \theta$$

$$Z' = Z$$

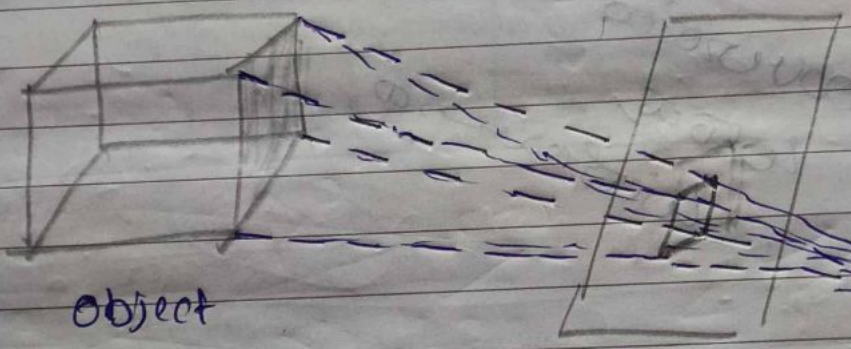
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



object  
object

View plane

Parallel projection



object

View plane

center of  
projection  
(Projection  
point)

perspective, projection  
plane